Statistical Methods
Unit III : Non-Parametric Tests and ANOVA

Mann–Whitney, Kruskal–Wallis, Sign and Runs Tests;
One-Way, Two-Way and Latin Square Designs

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Part 1
Rank-Based Tests: Mann–Whitney U and Kruskal–Wallis H

Recap: Why Non-Parametric Tests?

Definition

Non-parametric tests are procedures for testing hypotheses about population distributions that do *not* depend on population parameters such as the mean or variance.

They are used when

- the data are non-normal, or
- meaningful sample statistics (mean, variance) cannot be calculated or are not appropriate, or
- the sample size is small ($n < 50$), or
- the data are only *ordinal* (ranks, scores, grades).

The Two Rank-Sum Tests

	Mann–Whitney U	**Kruskal–Wallis** H
Samples	**two** independent samples	**three or more** independent samples
Parametric analogue	independent-samples t-test	one-way ANOVA
Statistic	U	H

Common idea. Pool *all* observations, rank them from smallest to largest, and compare the *rank sums* of the groups.

1. The Mann–Whitney U Test

Motivation. The sign test uses only the *signs* of paired differences and ignores the magnitudes — this discards useful information.

The **Mann–Whitney U test** is a *rank sum test* that partially avoids this loss by using the *relative magnitudes* (ranks) of the observations. It determines whether two independent samples have come from **identical populations**.

Assumptions and Hypotheses

1. The two samples are **independent** random samples.

2. The observations are at least **ordinal** (rankable), or continuous.
3. The two populations have **similar shapes**, so that a difference is interpretable as a difference in location (median).

H_0 : the two samples come from identical populations,
H_1 : the two samples do *not* come from identical populations.

The Formula

Let the two independent samples have sizes n_1 and n_2. Combine all $n_1 + n_2$ observations and rank them from 1 (smallest) to $n_1 + n_2$ (largest), giving *average ranks* to ties. Let R_1, R_2 be the rank sums of samples 1 and 2.

Mann–Whitney U Test: Statistic and Normal Approximation

$$U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1, \qquad U_2 = n_1 n_2 - U_1, \qquad U = \min(U_1,\, U_2),$$

$$\mu_U = \frac{n_1 n_2}{2}, \qquad \sigma_U = \sqrt{\frac{n_1 n_2\,(n_1 + n_2 + 1)}{12}}, \qquad Z = \frac{U - \mu_U}{\sigma_U}.$$

At 5% LOS (two-tailed), reject H_0 if $|Z| \geq 1.96$. For small samples, reject H_0 if $U \leq U_\alpha(n_1, n_2)$ from the Mann–Whitney table.

Remarks.

- **One formula only.** U_1 is computed from the formula above; U_2 always follows from $U_2 = n_1 n_2 - U_1$.
- **Built-in check:** $U_1 + U_2 = n_1 n_2$.
- **Rank-sum check:** $R_1 + R_2 = \dfrac{(n_1 + n_2)(n_1 + n_2 + 1)}{2}$.
- U is always the *smaller* of the two; hence $0 \leq U \leq \dfrac{n_1 n_2}{2}$.
- **Validity of the normal approximation:** use it when $n_1 \geq 8$ and $n_2 \geq 8$, *or* when the smaller sample size is at least 3 and $n_1 n_2 \geq 20$.
- In the small-sample table, note the direction: a *small* U is significant.

Working Procedure

1. State H_0 and H_1.
2. Pool the $n_1 + n_2$ observations; rank $1, 2, \ldots, n_1 + n_2$ (average ranks for ties).
3. Compute R_1 and R_2; verify $R_1 + R_2 = \frac{(n_1+n_2)(n_1+n_2+1)}{2}$.
4. Compute $U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1$, $U_2 = n_1 n_2 - U_1$, and $U = \min(U_1, U_2)$.

5. Compute μ_U, σ_U and $Z = (U - \mu_U)/\sigma_U$.
6. Compare with the critical value and conclude.

Worked Examples

Example 1: Suitability Scores

Twenty-three applicants are interviewed by three administrators and rated on a 5-point scale; the sum of the three ratings is the *suitability score*. Twelve applicants have completed at least two years of college. Use the Mann–Whitney U test at 5% LOS to test whether scores differ between **Group A** (less than two years of college) and **Group B** (at least two years of college).

Group A ($n_1 = 11$)	7	11	9	4	8	6	12	11	9	10	11	
Group B ($n_2 = 12$)	8	9	13	14	11	10	12	14	13	9	10	8

Solution.

Step 1: Hypotheses.

H_0 : the two groups come from identical populations,
H_1 : the two groups do not come from identical populations.

Step 2: Ranking (with ties). Tied scores share the *average* of the positions they occupy.

Score	Count in A	Count in B	Rank positions	Average rank
4	1	0	1	1.0
6	1	0	2	2.0
7	1	0	3	3.0
8	1	2	4–6	5.0
9	2	2	7–10	8.5
10	1	2	11–13	12.0
11	3	1	14–17	15.5
12	1	1	18–19	18.5
13	0	2	20–21	20.5
14	0	2	22–23	22.5

Step 3: Rank sums.

$$\begin{aligned} R_1 &= 1 + 2 + 3 + 5 + 2(8.5) + 12 + 3(15.5) + 18.5 \\ &= 1 + 2 + 3 + 5 + 17 + 12 + 46.5 + 18.5 = 105, \\ R_2 &= 2(5) + 2(8.5) + 2(12) + 15.5 + 18.5 + 2(20.5) + 2(22.5) \\ &= 10 + 17 + 24 + 15.5 + 18.5 + 41 + 45 = 171, \end{aligned}$$

Check: $R_1 + R_2 = 276 = \dfrac{23 \times 24}{2}$.

Step 4: Compute U. With $n_1 = 11$, $n_2 = 12$, $R_1 = 105$:

$$\begin{aligned} U_1 &= n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1 = 132 + 66 - 105 = 93, \\ U_2 &= n_1 n_2 - U_1 = 132 - 93 = 39, \\ U &= \min(U_1, U_2) = \min(93, 39) = 39. \end{aligned}$$

Step 5: Test statistic.

$$\begin{aligned} \mu_U &= \frac{n_1 n_2}{2} = \frac{132}{2} = 66, \\ \sigma_U &= \sqrt{\frac{n_1 n_2 (n_1 + n_2 + 1)}{12}} = \sqrt{\frac{11 \times 12 \times 24}{12}} = \sqrt{264} \approx 16.25, \\ Z &= \frac{U - \mu_U}{\sigma_U} = \frac{39 - 66}{16.25} = \frac{-27}{16.25} \approx -1.66. \end{aligned}$$

Step 6: Decision. At 5% LOS (two-tailed) $z_{0.025} = 1.96$. Since $|Z| \approx 1.66 < 1.96$, we *do not reject* H_0.

Conclusion. *At the 5% level of significance there is no significant difference between the suitability scores of the two groups. Educational background (less than two years vs. at least two years of college) does not appear to affect the suitability score.*

Practice Question M1 (Question Bank, 10 marks)

Two different teaching methods are compared by testing students' performance scores. Use the Mann–Whitney U test to determine whether there is a significant difference between the two methods at $\alpha = 0.05$.

Method A: 78, 85, 92, 88, 91, 87, 94
Method B: 82, 76, 89, 84, 86, 90, 83, 79

Solution.

Step 1: Hypotheses.

H_0 : No difference in median performance between the two methods,
H_1 : A significant difference exists between the two methods.

Sample sizes: $n_1 = n_A = 7$, $n_2 = n_B = 8$, $n_1 + n_2 = 15$.

Step 2: Combine and rank. Combined data in ascending order (no ties):

76, 78, 79, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 94.

Value	76	78	79	82	83	84	85	86	87	88	89	90	91	92	94
Rank	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Method	B	A	B	B	B	B	A	B	A	A	B	B	A	A	A

Step 3: Rank sums.

$$R_1 = R_A = 2 + 7 + 9 + 10 + 13 + 14 + 15 = 70,$$
$$R_2 = R_B = 1 + 3 + 4 + 5 + 6 + 8 + 11 + 12 = 50,$$
$$\text{Check: } R_1 + R_2 = 120 = \frac{15 \times 16}{2}.$$

Step 4: Compute U (same formula). With $n_1 = 7$, $n_2 = 8$, $R_1 = 70$:

$$U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1 = 56 + 28 - 70 = 14,$$
$$U_2 = n_1 n_2 - U_1 = 56 - 14 = 42,$$
$$U = \min(U_1, U_2) = \min(14, 42) = 14.$$

Check: $U_1 + U_2 = 14 + 42 = 56 = n_1 n_2$.

Step 5: Test statistic. Since the smaller sample size is $7 \geq 3$ and $n_1 n_2 = 56 \geq 20$, the normal approximation may be used.

$$\mu_U = \frac{n_1 n_2}{2} = \frac{56}{2} = 28,$$
$$\sigma_U = \sqrt{\frac{n_1 n_2 (n_1 + n_2 + 1)}{12}} = \sqrt{\frac{7 \times 8 \times 16}{12}} = \sqrt{74.67} \approx 8.64,$$
$$Z = \frac{U - \mu_U}{\sigma_U} = \frac{14 - 28}{8.64} = \frac{-14}{8.64} \approx -1.62.$$

Step 6: Decision. *Normal approximation:* $z_{0.025} = 1.96$ and $|Z| \approx 1.62 < 1.96$, so we fail to reject H_0.
Critical-value table (cross-check): for $n_1 = 7$, $n_2 = 8$ at $\alpha = 0.05$ (two-tailed), $U_{0.05} = 10$. Since $U = 14 > 10$, we again fail to reject H_0. Both routes agree.

Conclusion. *There is no significant difference between the two teaching methods at the 5% level of significance. The observed advantage of Method A is within the range attributable to sampling variation.*

2. The Kruskal–Wallis H Test

Definition

The **Kruskal–Wallis H test** is the non-parametric alternative to one-way analysis of variance (ANOVA). It tests whether **three or more** independent samples come from the same population, when normality or equal variances cannot be assumed. All observations from the k groups are combined and ranked from lowest to highest *as if they were one single sample.*

When to prefer H over one-way ANOVA:

- data are **not normally distributed**;
- variances are **unequal** across the groups;
- data are **ordinal** rather than interval/ratio;
- sample sizes are **small**, so normality is doubtful.

Hypotheses.

$$H_0: \text{ all } k \text{ samples come from the same population,}$$
$$H_1: \text{ at least one sample differs from the others.}$$

The Formula

Let k = number of groups, n_i = size of group i, $N = n_1 + n_2 + \cdots + n_k$, and R_i = sum of ranks of group i.

Kruskal–Wallis H Test: Statistic and Decision Rule

$$H = \frac{12}{N(N+1)} \sum_{i=1}^{k} \frac{R_i^2}{n_i} - 3(N+1), \qquad v = k-1 \text{ degrees of freedom.}$$

Reject H_0 if $H > \chi^2_{\alpha,\,k-1}$; otherwise accept H_0. For $\alpha = 0.05$, $k = 3$: $\chi^2_{0.05,2} = 5.991$.

Remarks.

- **One formula only** — the same expression is used for every problem, with $k = 3$ here but any $k \geq 3$ allowed.
- **Rank-sum check:** $\displaystyle\sum_{i=1}^{k} R_i = \frac{N(N+1)}{2}$.
- $H \geq 0$; a large H means the rank sums are far apart.
- For sufficiently large samples H is approximately χ^2 distributed with $k-1$ d.f.

Working Procedure

1. State H_0 and H_1.
2. Pool all N observations and rank $1, \ldots, N$ (average ranks for ties).
3. Find n_i and R_i for each group; verify $\sum R_i = \frac{N(N+1)}{2}$.
4. Compute each $\dfrac{R_i^2}{n_i}$ and their sum.
5. Compute $H = \dfrac{12}{N(N+1)} \sum \dfrac{R_i^2}{n_i} - 3(N+1)$.
6. Compare H with $\chi^2_{\alpha,k-1}$ and conclude.

Worked Examples

Example 2: Inspection Training Methods

A company's trainees are randomly assigned to groups and taught an industrial inspection procedure by three different methods. At the end of the training they are tested for inspection performance quality.

Method A: 80, 83, 79, 85, 90, 68
Method B: 82, 84, 60, 72, 86, 67, 91
Method C: 93, 65, 77, 78, 88

Use the Kruskal–Wallis H test at 5% LOS to determine whether the three methods are equally effective.

Solution.

Step 1: Hypotheses.

H_0 : all three methods are equally effective (same distribution),
H_1 : at least one method differs in effectiveness.

Sizes: $n_1 = 6$, $n_2 = 7$, $n_3 = 5$, $N = 18$, $k = 3$.

Step 2: Ranking all 18 scores.

Score	60	65	67	68	72	77	78	79	80	82	83	84	85	86	88	90	91	93
Rank	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Method	B	C	B	A	B	C	C	A	A	B	A	B	A	B	C	A	B	C

Step 3: Rank sums.

$$R_1 = R_A = 4 + 8 + 9 + 11 + 13 + 16 = 61,$$
$$R_2 = R_B = 1 + 3 + 5 + 10 + 12 + 14 + 17 = 62,$$
$$R_3 = R_C = 2 + 6 + 7 + 15 + 18 = 48,$$

Check: $R_1 + R_2 + R_3 = 171 = \dfrac{18 \times 19}{2}$.

Step 4: Compute H.

$$\frac{61^2}{6} = \frac{3721}{6} \approx 620.17, \qquad \frac{62^2}{7} = \frac{3844}{7} \approx 549.14, \qquad \frac{48^2}{5} = \frac{2304}{5} = 460.80.$$

$$\begin{aligned} H &= \frac{12}{N(N+1)}\left(\frac{R_1^2}{n_1} + \frac{R_2^2}{n_2} + \frac{R_3^2}{n_3}\right) - 3(N+1) \\ &\approx \frac{12}{18 \times 19}(620.17 + 549.14 + 460.80) - 3(19) \\ &= \frac{12}{342}(1630.11) - 57 \approx 57.20 - 57 \approx 0.20. \end{aligned}$$

Step 5: Decision. Degrees of freedom $v = k - 1 = 2$; from the χ^2 table, $\chi^2_{0.05,\,2} = 5.991$. Since $H \approx 0.20 < 5.991$, we *do not reject* H_0.

Conclusion. *At the 5% level of significance the Kruskal–Wallis test indicates no significant difference among the three training methods; they may be regarded as equally effective.*

Practice Question K1 (Question Bank, 10 marks)

Three different fertilizers are tested on crop yields. Use the Kruskal–Wallis H test to determine whether there is a significant difference between the fertilizers at $\alpha = 0.05$.

Fertilizer A: 45, 52, 48, 51, 46
Fertilizer B: 41, 38, 44, 42, 39, 43
Fertilizer C: 55, 58, 53, 56, 60

Solution.

Step 1: Hypotheses.

H_0 : no difference in median yields between the fertilizers,
H_1 : at least one fertilizer has a different median yield.

Sizes: $n_1 = n_A = 5$, $n_2 = n_B = 6$, $n_3 = n_C = 5$, $N = 16$, $k = 3$.

Step 2: Combine and rank. Combined ordered data (no ties):

Yield	38	39	41	42	43	44	45	46	48	51	52	53	55	56	58	60
Rank	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Fertilizer	B	B	B	B	B	B	A	A	A	A	A	C	C	C	C	C

Step 3: Rank sums.

$$\begin{aligned} R_1 = R_A &= 7 + 8 + 9 + 10 + 11 = 45, \\ R_2 = R_B &= 1 + 2 + 3 + 4 + 5 + 6 = 21, \\ R_3 = R_C &= 12 + 13 + 14 + 15 + 16 = 70, \end{aligned}$$

Check: $R_1 + R_2 + R_3 = 136 = \dfrac{16 \times 17}{2}$.

Step 4: Compute H (same formula).

$$\frac{45^2}{5} = \frac{2025}{5} = 405, \qquad \frac{21^2}{6} = \frac{441}{6} = 73.5, \qquad \frac{70^2}{5} = \frac{4900}{5} = 980.$$

$$\begin{aligned} H &= \frac{12}{16 \times 17}(405 + 73.5 + 980) - 3(17) \\ &= \frac{12}{272}(1458.5) - 51 = \frac{17502}{272} - 51 \approx 64.35 - 51 \approx 13.35. \end{aligned}$$

Step 5: Decision. Degrees of freedom $v = k - 1 = 2$; $\chi^2_{0.05,\,2} = 5.991$. Since $H \approx 13.35 > 5.991$, we **reject** H_0.

Conclusion. *There is a significant difference between the fertilizers' effects on crop yields at the 5% level of significance. Inspection of the rank sums indicates that Fertilizer C ($R_C = 70$ from only 5 observations) is the most effective, while Fertilizer B ($R_B = 21$ from 6 observations) gives the lowest yields.*

Summary: Mann–Whitney and Kruskal–Wallis

	Mann–Whitney U	**Kruskal–Wallis H**
Samples compared	2	$k \geq 3$
Parametric analogue	Independent t-test	One-way ANOVA
Statistic	$U = \min(U_1, U_2)$	H
Distribution used	Normal (Z), or U-table	χ^2 with $k-1$ d.f.
Reject H_0 when	$\lvert Z\rvert \geq 1.96$ or $U \leq U_\alpha$	$H > \chi^2_{\alpha,k-1}$

Problem	Test	Statistic	Critical	Decision
Example 1: Suitability scores	U	$U = 39$, $Z = -1.66$	1.96	Accept H_0
M1: Teaching methods	U	$U = 14$, $Z = -1.62$	1.96	Accept H_0
Example 2: Training methods	H	$H = 0.20$	5.991	Accept H_0
K1: Fertilizers	H	$H = 13.35$	5.991	**Reject** H_0

Only the fertilizer data provide evidence of a real difference between the groups.

Part 2
The Sign Test and the One-Sample Runs Test

3. The Sign Test (Wilcoxon Signed-Rank Test)

The **Sign Test** (in its Wilcoxon signed-rank form) is a simple non-parametric test for a median, or for the median of paired differences. It uses the sign (+ or −) of each difference together with the *rank* of its magnitude.

For each observation, compute the difference d_i (from the hypothesised median, or between the pair), discard any $d_i = 0$, and rank the remaining $|d_i|$ from smallest to largest (tied values receive the average of the ranks they occupy). Let n be the number of non-zero differences, T^+ the sum of ranks of positive differences and T^- the sum of ranks of negative differences.

Sign Test (Wilcoxon Signed-Rank): Test Statistic and Decision Rule

$$T = \min(T^+, T^-), \qquad T^+ + T^- = \frac{n(n+1)}{2},$$

$$\mu_T = \frac{n(n+1)}{4}, \qquad \sigma_T = \sqrt{\frac{n(n+1)(2n+1)}{24}},$$

$$Z = \frac{T - \mu_T}{\sigma_T}.$$

At 5% LOS, reject H_0 if $|Z| > 1.96$ (two-tailed) or $Z < -1.645$ (one-tailed). Equivalently, reject H_0 if $T \leq T_\alpha$, the critical value from Wilcoxon tables.

Types of Sign Test

- **One-sample sign test:** We have a single sample of size n. Each observation is compared with a hypothesised median.
- **Paired-sample sign test:** Data are collected in pairs (e.g., before and after some treatment). Each pair (x_i, y_i) is replaced by
  - a plus sign + if $x_i > y_i$,
  - a minus sign − if $x_i < y_i$,
  - discarded if $x_i = y_i$.

Worked Examples

Example 1

A typing school claims that in a six-week intensive course, it can train students to type, on the average, at least 60 words per minute. A random sample of 15 graduates is given a typing test, and the median number of words per minute typed by each of these students is as follows:

Student	A	B	C	D	E
Words per minute	81	76	53	71	66
Student	F	G	H	I	J
Words per minute	59	88	73	80	66
Student	K	L	M	N	O
Words per minute	58	70	60	56	55

Test the hypothesis that the median typing speed of graduates is at least 60 words per minute at 5% LOS using the sign test.

Solution. We test whether the population median typing speed is at least 60 words per minute, using the signed-rank statistic $T = \min(T^+, T^-)$ and its normal approximation (see Practice Question S2).

Step 1: Hypotheses.

$$H_0 : \text{Median typing speed } M_d = 60,$$
$$H_1 : \text{Median typing speed } M_d > 60 \quad \text{(one-tailed)}.$$

Step 2: Differences, signs and ranks. Compute $d_i = x_i - 60$, discard $d_i = 0$ (student M), and rank the remaining $|d_i|$ from smallest to largest. Tied values ($|d| = 6$) receive the average of the ranks they occupy, $(5 + 6)/2 = 5.5$.

Student	x_i	$d_i = x_i - 60$	$\lvert d_i \rvert$	Signed rank
A	81	+21	21	+13
B	76	+16	16	+11
C	53	−7	7	−7
D	71	+11	11	+9
E	66	+6	6	+5.5
F	59	−1	1	−1
G	88	+28	28	+14
H	73	+13	13	+10
I	80	+20	20	+12
J	66	+6	6	+5.5
K	58	−2	2	−2
L	70	+10	10	+8
M	60	0	–	discarded
N	56	−4	4	−3
O	55	−5	5	−4

Number of non-zero differences: $n = 14$.

Step 3: Rank sums and test statistic.

$$\begin{aligned} T^+ &= 13 + 11 + 9 + 5.5 + 14 + 10 + 12 + 5.5 + 8 = 88, \\ T^- &= 7 + 1 + 2 + 3 + 4 = 17, \\ \text{Check: } T^+ + T^- &= 105 = \frac{n(n+1)}{2} = \frac{14 \times 15}{2}, \\ T &= \min(T^+, T^-) = \min(88, 17) = 17. \end{aligned}$$

Step 4: Normal approximation.

$$\begin{aligned} \mu_T &= \frac{n(n+1)}{4} = \frac{14 \times 15}{4} = 52.5, \\ \sigma_T &= \sqrt{\frac{n(n+1)(2n+1)}{24}} = \sqrt{\frac{14 \times 15 \times 29}{24}} = \sqrt{253.75} \approx 15.93, \\ Z &= \frac{T - \mu_T}{\sigma_T} = \frac{17 - 52.5}{15.93} = \frac{-35.5}{15.93} \approx -2.23. \end{aligned}$$

Step 5: Decision. For a one-tailed test at 5% LOS, the critical value is $z_{0.05} = -1.645$. Since $Z \approx -2.23 < -1.645$, we *reject* H_0. (Equivalently, from Wilcoxon tables for $n = 14$, one-tailed $\alpha = 0.05$: $T_{0.05} = 25$, and $T = 17 < 25$.)

Conclusion. *At the 5% level of significance, the median typing speed of graduates is significantly greater than 60 words per minute. The data support the typing school's claim.*

Example 2

Use the sign test to determine whether there is a difference between the number of days until the collection of an account receivable before and after a new collection policy.

Before	30	28	34	35	40	42	33	38	34	45	28	27	25	41	36
After	32	29	33	32	37	43	40	41	37	44	27	33	30	38	36

Solution. We apply the signed-rank statistic $T = \min(T^+, T^-)$ and its normal approximation (see Practice Question S2) to the paired differences.

Step 1: Hypotheses.

$$\begin{aligned} H_0 &: \text{Median difference} = 0 \quad \text{(no change in collection time)}, \\ H_1 &: \text{Median difference} \neq 0 \quad \text{(two-tailed)}. \end{aligned}$$

Step 2: Differences, signs and ranks. Take $d_i = \text{Before}_i - \text{After}_i$, discard $d_i = 0$ (pair 15), and rank the remaining $|d_i|$. Ties receive average ranks: $|d| = 1$ occurs 5 times (ranks 1–5, average 3) and $|d| = 3$ occurs 5 times (ranks 7–11, average 9).

Pair	Before	After	d_i	$\lvert d_i \rvert$	Signed rank
1	30	32	-2	2	-6
2	28	29	-1	1	-3
3	34	33	$+1$	1	$+3$
4	35	32	$+3$	3	$+9$
5	40	37	$+3$	3	$+9$
6	42	43	-1	1	-3
7	33	40	-7	7	-14
8	38	41	-3	3	-9
9	34	37	-3	3	-9
10	45	44	$+1$	1	$+3$
11	28	27	$+1$	1	$+3$
12	27	33	-6	6	-13
13	25	30	-5	5	-12
14	41	38	$+3$	3	$+9$
15	36	36	0	–	discarded

Number of non-zero differences: $n = 14$.

Step 3: Rank sums and test statistic.

$$\begin{aligned} T^+ &= 3+9+9+3+3+9 = 36, \\ T^- &= 6+3+3+14+9+9+13+12 = 69, \\ \text{Check: } T^+ + T^- &= 105 = \frac{14\times 15}{2}, \\ T &= \min(T^+, T^-) = \min(36, 69) = 36. \end{aligned}$$

Step 4: Normal approximation. As in Example 1 (same $n = 14$), $\mu_T = 52.5$ and $\sigma_T \approx 15.93$. Hence

$$Z = \frac{T - \mu_T}{\sigma_T} = \frac{36 - 52.5}{15.93} = \frac{-16.5}{15.93} \approx -1.04.$$

Step 5: Decision. For a two-tailed test at 5% LOS, the critical values are ± 1.96. Since $|Z| \approx 1.04 < 1.96$, we *do not reject* H_0. (Equivalently, from Wilcoxon tables for $n = 14$, two-tailed $\alpha = 0.05$: $T_{0.05} = 21$, and $T = 36 > 21$.)

Conclusion. *There is no significant difference between the collection times before and after the new policy, at the 5% level of significance.*

Practice Questions: Sign Test / Wilcoxon Signed-Rank Test

Practice Question S1 (2 marks)

State the null and alternative hypotheses for the Wilcoxon signed-rank test.

Solution. Let d denote the paired difference (e.g. After − Before).

$$H_0 : \text{Median difference is zero (no significant difference)},$$
$$H_1 : \text{Median difference is not zero (a significant difference exists)}.$$

Alternatively, the hypotheses may be written as $H_0 : \mu_d = 0$ and $H_1 : \mu_d \neq 0$, where μ_d denotes the median (or mean, in some contexts) of the differences.

Practice Question S2 (2 marks)

What is the test statistic formula for the Wilcoxon signed-rank test?

Solution. Let T^+ be the sum of ranks of positive differences and T^- the sum of ranks of negative differences, and let n be the number of non-zero differences.

Wilcoxon Signed-Rank Statistic

$$T = \min(T^+, T^-),$$
$$Z = \frac{T - \dfrac{n(n+1)}{4}}{\sqrt{\dfrac{n(n+1)(2n+1)}{24}}} \quad \text{(for large samples)}.$$

Practice Question S3 (10 marks)

Ten subjects were measured before and after a training program. Use the Wilcoxon signed-rank test to determine if the training program is effective at $\alpha = 0.05$.

Subject	1	2	3	4	5	6	7	8	9	10
Before	65	78	82	74	69	85	71	79	77	73
After	72	81	85	78	75	87	76	84	80	76

Solution. Step 1: Hypotheses

$$H_0 : \text{Median difference} = 0 \quad \text{(no improvement)},$$
$$H_1 : \text{Median difference} > 0 \quad \text{(improvement exists)}.$$

Step 2: Differences and ranks ($d = \text{After} - \text{Before}$)

Ordered $|d|$ values: $2,\ 3, 3, 3, 3,\ 4,\ 5, 5,\ 6,\ 7$. Ties receive the average of the ranks they occupy: $|d| = 3$ occupies ranks 2–5 (average 3.5), and $|d| = 5$ occupies ranks 7–8 (average 7.5).

Subject	Before	After	Difference d	$\lvert d\rvert$	Rank
1	65	72	+7	7	10
2	78	81	+3	3	3.5
3	82	85	+3	3	3.5
4	74	78	+4	4	6
5	69	75	+6	6	9
6	85	87	+2	2	1
7	71	76	+5	5	7.5
8	79	84	+5	5	7.5
9	77	80	+3	3	3.5
10	73	76	+3	3	3.5

Step 3: Rank sums

All differences are positive, so

$$T^+ = 10 + 3.5 + 3.5 + 6 + 9 + 1 + 7.5 + 7.5 + 3.5 + 3.5 = 55 \quad \left(= \tfrac{10\times 11}{2}\right),$$
$$T^- = 0, \qquad T = \min(T^+, T^-) = 0.$$

Step 4: Critical value

For $n = 10$ non-zero pairs, at $\alpha = 0.05$ (one-tailed), the critical value from Wilcoxon tables is $T_{0.05} = 10$.

Step 5: Decision

Since $T = 0 < 10$, we reject H_0.

Conclusion. *The training program is significantly effective at improving performance at* $\alpha = 0.05$.

4. One-Sample Runs Test

The **one-sample runs test** is used to judge the randomness of a sequence based on the order in which observations occur.

A *run* is a succession of identical symbols that is preceded and followed by a different symbol (or by no symbol at all).

Let n_1 and n_2 be the numbers of two types of symbols (e.g. M and W), and let r be the total number of runs. For large samples (typically $n_1 > 20$ or $n_2 > 20$), r is approximately normal with the following mean and variance.

Runs Test: Mean, Variance and Test Statistic

$$\mu_r = \frac{2n_1n_2}{n_1+n_2} + 1,$$
$$\sigma_r^2 = \frac{2n_1n_2\left(2n_1n_2 - n_1 - n_2\right)}{(n_1+n_2)^2(n_1+n_2-1)},$$
$$Z = \frac{r-\mu_r}{\sigma_r}.$$

At 5% LOS (two-tailed), reject H_0 (randomness) if $|Z| > 1.96$.

Worked Example

Example 3

An arrangement of 25 men (M) and 15 women (W) lined up to purchase tickets for a premier show is observed as follows:

M M M M W W M M M W M M W W W M M M W W
M M M M W M M W W M M M W W W M W M M M

Test whether the arrangement is random at the 5% LOS.

Solution. Marking the runs in the queue (each run separated by "|"):

MMMM | WW | MMM | W | MM | WWW | MMM | WW |
MMMM | W | MM | WW | MMM | WWW | M | W | MMM

There are 9 runs of M and 8 runs of W. Hence $n_1 = 25$ (men), $n_2 = 15$ (women), and $r = 9 + 8 = 17$.

Mean of r.

$$\mu_r = \frac{2n_1n_2}{n_1+n_2} + 1 = \frac{2\times 25\times 15}{40} + 1 = \frac{750}{40} + 1 = 18.75 + 1 = 19.75.$$

Variance of r.

$$\sigma_r^2 = \frac{2n_1n_2\big(2n_1n_2 - n_1 - n_2\big)}{(n_1+n_2)^2(n_1+n_2-1)} = \frac{750 \times (750-40)}{1600 \times 39} = \frac{750 \times 710}{62400} \approx 8.534,$$
$$\sigma_r \approx \sqrt{8.534} \approx 2.92.$$

Test statistic.

$$Z = \frac{r - \mu_r}{\sigma_r} = \frac{17 - 19.75}{2.92} \approx \frac{-2.75}{2.92} \approx -0.94.$$

At 5% LOS for a two-sided test, $z_{0.025} = 1.96$. Since $|Z| \approx 0.94 < 1.96$, we do not reject the null hypothesis of randomness.

Conclusion. *There is no evidence at the 5% LOS to suggest that the arrangement is not random.*

Practice Questions: One-Sample Runs Test

Practice Question R1 (2 marks)

What does a "run" represent in the runs test for randomness?

Solution. A **run** is a sequence of consecutive identical symbols or values in a series. For example, in the sequence

A A A B B A

there are 3 runs: AAA, BB, and A at the end. The runs test examines whether the observed number of runs is compatible with randomness or indicates a pattern in the data sequence.

Practice Question R2 (2 marks)

In a runs test, if we have 15 runs with $n_1 = 12$ and $n_2 = 8$, calculate the expected number of runs.

Solution.

$$E(R) = \frac{2n_1n_2}{n_1+n_2} + 1 = \frac{2 \cdot 12 \cdot 8}{12+8} + 1 = \frac{192}{20} + 1 = 9.6 + 1 = 10.6.$$

Practice Question R3 (2 marks)

State the decision rule for the runs test when testing for randomness at $\alpha = 0.05$.

Solution.

- If the observed number of runs is significantly low or significantly high compared with the expected number of runs, reject H_0 (randomness).
- For large samples, compute $Z = \dfrac{R - E(R)}{\sigma_R}$ and reject H_0 if $|Z| > 1.96$ ($\alpha = 0.05$, two-tailed).
- Too few runs suggest clustering; too many runs suggest systematic alternation.

Practice Question R4 (10 marks)

A quality control manager wants to test if a production process is random. The following sequence represents defective (D) and non-defective (N) items in order of production. Test for randomness at $\alpha = 0.05$.

Sequence:
N N N D D N N D D D N N N D N D D N N N D D D N N

Solution. Step 1: Hypotheses

$$H_0 : \text{The sequence is random,}$$
$$H_1 : \text{The sequence is not random.}$$

Step 2: Count elements and runs

Identify runs in the sequence (separating each run by "|"):

$$\text{NNN} \,|\, \text{DD} \,|\, \text{NN} \,|\, \text{DDD} \,|\, \text{NNN} \,|\, \text{D} \,|\, \text{N} \,|\, \text{DD} \,|\, \text{NNN} \,|\, \text{DDD} \,|\, \text{NN}.$$

Number of runs: $R = 11$.

Number of non-defectives: $n_1 = 3 + 2 + 3 + 1 + 3 + 2 = 14$; number of defectives: $n_2 = 2 + 3 + 1 + 2 + 3 = 11$; total observations: $N = n_1 + n_2 = 25$.

Step 3: Expected number of runs

$$E(R) = \frac{2n_1n_2}{n_1 + n_2} + 1 = \frac{2 \cdot 14 \cdot 11}{25} + 1 = \frac{308}{25} + 1 = 12.32 + 1 = 13.32.$$

Step 4: Standard deviation of runs

$$\sigma_R = \sqrt{\frac{2n_1n_2(2n_1n_2 - n_1 - n_2)}{(n_1 + n_2)^2(n_1 + n_2 - 1)}} = \sqrt{\frac{308\,(308 - 25)}{25^2 \cdot 24}}$$
$$= \sqrt{\frac{308 \cdot 283}{15000}} = \sqrt{\frac{87164}{15000}} = \sqrt{5.811} \approx 2.411.$$

Step 5: Test statistic

$$Z = \frac{R - E(R)}{\sigma_R} = \frac{11 - 13.32}{2.411} = \frac{-2.32}{2.411} \approx -0.962.$$

Step 6: Critical values

For $\alpha = 0.05$ (two-tailed), the critical values are $Z_{0.025} = \pm 1.96$.

Step 7: Decision

Since $|Z| = 0.962 < 1.96$, we fail to reject H_0.

Conclusion. *There is insufficient evidence to conclude that the production process is non-random. The sequence appears to be random at the 5% significance level.*

Part 3
Analysis of Variance (ANOVA): One-Way, Two-Way and Latin Square Designs

Introduction to ANOVA

Definition

ANOVA (Analysis of Variance) is a statistical method used to analyse the differences among group means in a sample. It helps determine whether there are any statistically significant differences between the means of three or more independent (unrelated) groups.
ANOVA works by comparing the variation between groups (due to treatment effects) with the variation within groups (due to random error). If the between-group variation is significantly larger than the within-group variation, we conclude that at least one group mean is different from the others.

Overview of Experimental Designs

There are four widely used designs in statistical experimentation. Each design controls variation differently and applies to different research settings.

1. Completely Randomised Design (CRD) – one-way ANOVA
2. Randomised Block Design (RBD) – two-way ANOVA
3. Latin Square Design (LSD) – three-way ANOVA
4. 2^2 Factorial Design – two-way ANOVA (balanced factorial structure)

2^2 Factorial Design

A 2^2 **Factorial Design** studies two independent factors, each at two levels (e.g. High–Low), giving $2 \times 2 = 4$ treatment combinations.

- Fully crossed design; measures both main (individual) effects and the interaction (joint) effect.
- Highly efficient and widely applicable.

Classifications of ANOVA

Definition

1. One-way ANOVA (single factor). Compares the means of three or more groups based on a single independent variable (factor). *Example:* effect of different diets on weight loss (factor: diet type).

2. Two-way ANOVA. Compares group means based on two independent variables (factors), assessing the main effect of each factor and, when the cells are replicated, the interaction between them. *Example:* effect of diet type and exercise routine on weight loss.

3. MANOVA (Multivariate ANOVA). Extends ANOVA to several dependent variables at once. *Example:* effect of a treatment on blood pressure, cholesterol and heart rate together.

Unified Notation and Solution Procedure

Notation used for all designs

The same symbols are used for every type of ANOVA problem in these notes.

Symbol	Meaning
x_{ij}	observation in row i and column j
R_i	row total (total of row i)
C_j	column total (total of column j)
T_k	treatment total (total of all cells carrying letter k; Latin square only)
T	grand total, $T = \sum R_i = \sum C_j$
N	total number of observations
$r,\ c,\ n$	number of rows, number of columns, order of a Latin square
n_j	number of observations in column j (one-way, unequal sizes)
CF	correction factor, $\mathrm{CF} = T^2/N$
SST	total sum of squares
SSR, SSC, SSTr	sum of squares between rows, columns, treatments
SSE	error (residual) sum of squares
$\nu_R,\ \nu_C,\ \nu_{Tr},\ \nu_E,\ \nu_T$	degrees of freedom of each source
MSR, MSC, MSTr, MSE	mean squares, $\mathrm{MS} = \mathrm{SS}/\nu$
$F_R,\ F_C,\ F_{Tr}$	variance ratios, $F = \mathrm{MS}/\mathrm{MSE}$

General solution procedure (all designs)

Step 1. State H_0 and H_1 for every source of variation.

Step 2. Write the given data as a table; compute the row totals R_i, the column totals C_j

(and, for a Latin square, the treatment totals T_k) and the grand total T.

Step 3. Write the table of squares x_{ij}^2 with its row and column totals to get $\sum\sum x_{ij}^2$.

Step 4. Correction factor $\text{CF} = T^2/N$.

Step 5. Sums of squares SST, SSR, SSC, SSTr and, by subtraction, SSE.

Step 6. Degrees of freedom and mean squares $\text{MS} = \text{SS}/\nu$.

Step 7. F-ratios $F = \text{MS}/\text{MSE}$.

Step 8. ANOVA table; compare each F_{cal} with $F_{0.05}(\nu, \nu_E)$ and conclude.

If $F_{\text{cal}} > F_{\text{tab}}$, H_0 is rejected (the effect is significant); otherwise H_0 is accepted. An F-ratio less than 1 is automatically smaller than any table value, so that effect is not significant.

Summary of the three designs

Design	Sources of variation	Error sum of squares	ν_E
One-way (CRD)	Columns (treatments)	$\text{SSE} = \text{SST} - \text{SSC}$	$N - c$
Two-way (RBD)	Rows, Columns	$\text{SSE} = \text{SST} - \text{SSR} - \text{SSC}$	$(r-1)(c-1)$
Latin square (LSD)	Rows, Columns, Treatments	$\text{SSE} = \text{SST} - \text{SSR} - \text{SSC} - \text{SSTr}$	$(n-1)(n-2)$

In every design: $\text{CF} = \dfrac{T^2}{N}$, $\text{SST} = \sum\sum x_{ij}^2 - \text{CF}$, $\nu_T = N - 1$.

One-Way ANOVA (Completely Randomised Design)

Completely Randomised Design (CRD)

A **Completely Randomised Design** is the simplest experimental design: experimental units are randomly assigned to treatments without any blocking. It is appropriate when all experimental units are homogeneous, is easy to implement and analyse, but does not control variability arising from differences between units. The data are analysed by **one-way ANOVA**.

Hypotheses

Consider c treatments (groups) with means $\mu_1, \mu_2, \ldots, \mu_c$, placed as the c columns of the data table.

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_c \quad \text{(all treatment means are equal)},$$
$$H_1 : \text{at least one treatment mean is different.}$$

Sums of squares

Let C_j be the total and n_j the number of observations of treatment j, T the grand total and $N = \sum n_j$.

$$\text{CF} = \frac{T^2}{N}, \qquad \text{SST} = \sum_i \sum_j x_{ij}^2 - \text{CF},$$

$$\text{SSC} = \sum_j \frac{C_j^2}{n_j} - \text{CF} \quad \text{(between treatments)},$$

$$\text{SSE} = \text{SST} - \text{SSC} \quad \text{(within treatments / error)}.$$

(SSC and SSE are the "between" and "within" sums of squares, often written SSB and SSW.)

Mean squares and F-ratio

$$\text{MSC} = \frac{\text{SSC}}{\nu_C}, \quad \nu_C = c - 1; \qquad \text{MSE} = \frac{\text{SSE}}{\nu_E}, \quad \nu_E = N - c,$$

$$F_C = \frac{\text{MSC}}{\text{MSE}} \quad \text{with } (\nu_C, \nu_E) \text{ d.f.}$$

If $F_C > F_{0.05}(\nu_C, \nu_E)$, reject H_0.

Part I: One-Way ANOVA – Worked Examples

Example 1.1: Diet Plans

Question

You want to test if there is a significant difference in the mean weight loss for three different diet plans A, B and C. The data collected (in kg) are:

Diet A	3	4	5	6	2
Diet B	8	9	7	10	9
Diet C	5	6	7	8	7

Test at 5% level of significance using one-way ANOVA.

Answer

Step 1: Hypotheses

- $H_0^{(C)}$: there is no significant difference between the three diet plans;
  $H_1^{(C)}$: there is a significant difference between the three diet plans.

Step 2: Data table with row and column totals

Treatments are placed column-wise. Column totals C_j give the treatment totals; row totals R_i are used only as a check.

Obs.	Diet A	Diet B	Diet C	R_i (row total)
1	3	8	5	16
2	4	9	6	19
3	5	7	7	19
4	6	10	8	24
5	2	9	7	18
C_j (column total)	20	43	33	$T = 96$
n_j (no. of obs.)	5	5	5	$N = 15$

Check: $\sum R_i = \sum C_j = T = 96$; $c = 3$, $N = 15$.

Step 3: Table of squares x_{ij}^2

Squaring every observation:

Obs.	Diet A	Diet B	Diet C	$\sum_j x_{ij}^2$
1	9	64	25	98
2	16	81	36	133
3	25	49	49	123
4	36	100	64	200
5	4	81	49	134
$\sum_i x_{ij}^2$	90	375	223	$\sum\sum x_{ij}^2 = 688$

Step 4: Correction factor

$$\text{CF} = \frac{T^2}{N} = \frac{96^2}{15} = \frac{9216}{15} = 614.4$$

Step 5: Sums of squares

$$\text{SST} = \sum\sum x_{ij}^2 - \text{CF} = 688 - 614.4 = 73.6$$

$$\begin{aligned}\text{SSC} &= \sum_j \frac{C_j^2}{n_j} - \text{CF} \\ &= \frac{20^2}{5} + \frac{43^2}{5} + \frac{33^2}{5} - \text{CF} \\ &= 80 + 369.8 + 217.8 - 614.4 \\ &= 667.6 - 614.4 = 53.2\end{aligned}$$

$$\text{SSE} = \text{SST} - \text{SSC} = 73.6 - 53.2 = 20.4$$

Step 6: Degrees of freedom and mean squares

$\nu_C = c - 1 = 2, \quad \nu_E = N - c = 15 - 3 = 12, \quad \nu_T = N - 1 = 14.$

$$\text{MSC} = \frac{\text{SSC}}{\nu_C} = \frac{53.2}{2} = 26.6$$

$$\text{MSE} = \frac{\text{SSE}}{\nu_E} = \frac{20.4}{12} = 1.7$$

Step 7: F-ratio

$$F_C = \frac{\text{MSC}}{\text{MSE}} = \frac{26.6}{1.7} \approx 15.65$$

with $(\nu_C, \nu_E) = (2, 12)$ d.f.

Step 8: ANOVA table and decision

Source	SS	d.f.	MS	F_{cal}	$F_{0.05}(\nu, \nu_E)$	Decision
Columns (treatments)	53.2	2	26.6	15.65	3.89	Significant
Error (within)	20.4	12	1.7	–	–	–
Total	73.6	14				

- $F_C = 15.65 > F_{0.05}(2, 12) = 3.89 \Rightarrow H_0^{(C)}$ is rejected: the difference between the three diet plans is **significant**.

Final Answer

The three diet plans do *not* give the same mean weight loss; at least one diet differs significantly from the others.

Example 1.2: Sales in Three Regions

Question

Sales data for three regions (North, South, West) are:

North	20	21	22	23	19
South	25	27	28	26	24
West	18	20	19	21	22

Test whether average sales differ between the three regions using one-way ANOVA at 5% LOS.

Answer

Step 1: Hypotheses

- $H_0^{(C)}$: there is no significant difference between the three regions;
  $H_1^{(C)}$: there is a significant difference between the three regions.

Step 2: Data table with row and column totals

Treatments are placed column-wise. Column totals C_j give the treatment totals; row totals R_i are used only as a check.

Obs.	North	South	West	R_i (row total)
1	20	25	18	63
2	21	27	20	68
3	22	28	19	69
4	23	26	21	70
5	19	24	22	65
C_j (column total)	105	130	100	$T = 335$
n_j (no. of obs.)	5	5	5	$N = 15$

Check: $\sum R_i = \sum C_j = T = 335$; $c = 3$, $N = 15$.

Step 3: Table of squares x_{ij}^2

Squaring every observation:

Obs.	North	South	West	$\sum_j x_{ij}^2$
1	400	625	324	1349
2	441	729	400	1570
3	484	784	361	1629
4	529	676	441	1646
5	361	576	484	1421
$\sum_i x_{ij}^2$	2215	3390	2010	$\sum\sum x_{ij}^2 = 7615$

Step 4: Correction factor

$$\text{CF} = \frac{T^2}{N} = \frac{335^2}{15} = \frac{112225}{15} \approx 7481.6667$$

Step 5: Sums of squares

$$\text{SST} = \sum\sum x_{ij}^2 - \text{CF} = 7615 - 7481.6667 \approx 133.3333$$

$$\begin{aligned}\text{SSC} &= \sum_j \frac{C_j^2}{n_j} - \text{CF} \\ &= \frac{105^2}{5} + \frac{130^2}{5} + \frac{100^2}{5} - \text{CF} \\ &= 2205 + 3380 + 2000 - 7481.6667 \\ &= 7585 - 7481.6667 \approx 103.3333\end{aligned}$$

$$\text{SSE} = \text{SST} - \text{SSC} = 133.3333 - 103.3333 = 30$$

Step 6: Degrees of freedom and mean squares

$\nu_C = c - 1 = 2, \quad \nu_E = N - c = 15 - 3 = 12, \quad \nu_T = N - 1 = 14.$

$$\text{MSC} = \frac{\text{SSC}}{\nu_C} = \frac{103.3333}{2} \approx 51.6667$$

$$\text{MSE} = \frac{\text{SSE}}{\nu_E} = \frac{30}{12} = 2.5$$

Step 7: F-ratio

$$F_C = \frac{\text{MSC}}{\text{MSE}} = \frac{51.6667}{2.5} \approx 20.67$$

with $(\nu_C, \nu_E) = (2, 12)$ d.f.

Step 8: ANOVA table and decision

Source	SS	d.f.	MS	F_{cal}	$F_{0.05}(\nu, \nu_E)$	Decision
Columns (treatments)	103.3333	2	51.6667	20.67	3.89	Significant
Error (within)	30	12	2.5	–	–	–
Total	133.3333	14				

- $F_C = 20.67 > F_{0.05}(2, 12) = 3.89 \Rightarrow H_0^{(C)}$ is rejected: the difference between the three regions is **significant**.

Final Answer

Average sales differ significantly between the three regions.

Example 1.3: Completely Randomised Design (Ten Plots)

Question

A completely randomised design experiment with 10 plots and 3 treatments gave the following results:

Plot No.	1	2	3	4	5	6	7	8	9	10
Treatment	A	B	C	A	C	C	A	B	A	B
Yield	5	4	3	7	5	1	3	4	1	7

Analyse the results for treatment effects using one-way ANOVA at 5% LOS.

Answer

Step 1: Hypotheses

- $H_0^{(C)}$: there is no significant difference between the three treatments;
  $H_1^{(C)}$: there is a significant difference between the three treatments.

Step 2: Data table with row and column totals

Rearranging the plots treatment-wise (A: plots 1, 4, 7, 9; B: plots 2, 8, 10; C: plots 3, 5, 6):
Treatments are placed column-wise. Column totals C_j give the treatment totals; row totals R_i are used only as a check.

Replicate	A	B	C	R_i (row total)
1	5	4	3	12
2	7	4	5	16
3	3	7	1	11
4	1	–	–	1
C_j (column total)	16	15	9	$T = 40$
n_j (no. of obs.)	4	3	3	$N = 10$

Check: $\sum R_i = \sum C_j = T = 40$; $c = 3$, $N = 10$.

Step 3: Table of squares x_{ij}^2
Squaring every observation:

Replicate	A	B	C	$\sum_j x_{ij}^2$
1	25	16	9	50
2	49	16	25	90
3	9	49	1	59
4	1	–	–	1
$\sum_i x_{ij}^2$	84	81	35	$\sum\sum x_{ij}^2 = 200$

Step 4: Correction factor

$$\text{CF} = \frac{T^2}{N} = \frac{40^2}{10} = \frac{1600}{10} = 160$$

Step 5: Sums of squares

$$\text{SST} = \sum\sum x_{ij}^2 - \text{CF} = 200 - 160 = 40$$

$$\begin{aligned}\text{SSC} &= \sum_j \frac{C_j^2}{n_j} - \text{CF} \\ &= \frac{16^2}{4} + \frac{15^2}{3} + \frac{9^2}{3} - \text{CF} \\ &= 64 + 75 + 27 - 160 \\ &= 166 - 160 = 6\end{aligned}$$

$$\text{SSE} = \text{SST} - \text{SSC} = 40 - 6 = 34$$

Step 6: Degrees of freedom and mean squares
$\nu_C = c - 1 = 2$, $\nu_E = N - c = 10 - 3 = 7$, $\nu_T = N - 1 = 9$.

$$\text{MSC} = \frac{\text{SSC}}{\nu_C} = \frac{6}{2} = 3$$

$$\text{MSE} = \frac{\text{SSE}}{\nu_E} = \frac{34}{7} \approx 4.8571$$

Step 7: F-ratio

$$F_C = \frac{\text{MSC}}{\text{MSE}} = \frac{3}{4.8571} \approx 0.62$$

with $(\nu_C, \nu_E) = (2, 7)$ d.f.

Step 8: ANOVA table and decision

Source	SS	d.f.	MS	F_{cal}	$F_{0.05}(\nu, \nu_E)$	Decision
Columns (treatments)	6	2	3	0.62	4.74	Not significant
Error (within)	34	7	4.8571	–	–	–
Total	40	9				

- $F_C = 0.62 < F_{0.05}(2, 7) = 4.74$ (in fact $F < 1$) $\Rightarrow H_0^{(C)}$ is accepted: the difference between the three treatments is **not significant**.

Final Answer

There is no significant difference between the treatment means; the three treatments may be regarded as equally effective.

Example 1.4: Lives of Electric Lamps

Question

The following table shows the lives (in hours) of four brands of electric lamps:

Brand A:	1610, 1610, 1650, 1680, 1700, 1720, 1800
Brand B:	1580, 1640, 1640, 1700, 1750
Brand C:	1460, 1550, 1600, 1620, 1640, 1660, 1740, 1820
Brand D:	1510, 1520, 1530, 1570, 1600, 1680

Perform an analysis of variance and test the homogeneity of the mean lives of the four brands of lamps at 5% LOS.

Answer

Step 1: Hypotheses

- $H_0^{(C)}$: there is no significant difference between the four brands;
  $H_1^{(C)}$: there is a significant difference between the four brands.

Step 2: Data table with row and column totals

The observations are large, so we work with the coded values $u_{ij} = \dfrac{x_{ij} - 1640}{10}$. A change of origin and scale multiplies every sum of squares by the same constant, so the F-ratio is unchanged. Coded data (brands column-wise):
Treatments are placed column-wise. Column totals C_j give the treatment totals; row totals R_i are used only as a check.

Obs.	A	B	C	D	R_i (row total)
1	-3	-6	-18	-13	-40
2	-3	0	-9	-12	-24
3	1	0	-4	-11	-14
4	4	6	-2	-7	1
5	6	11	0	-4	13
6	8	–	2	4	14
7	16	–	10	–	26
8	–	–	18	–	18
C_j (column total)	29	11	-3	-43	$T = -6$
n_j (no. of obs.)	7	5	8	6	$N = 26$

Check: $\sum R_i = \sum C_j = T = -6$; $c = 4$, $N = 26$.

Step 3: Table of squares x_{ij}^2
Squaring every observation:

Obs.	A	B	C	D	$\sum_j x_{ij}^2$
1	9	36	324	169	538
2	9	0	81	144	234
3	1	0	16	121	138
4	16	36	4	49	105
5	36	121	0	16	173
6	64	–	4	16	84
7	256	–	100	–	356
8	–	–	324	–	324
$\sum_i x_{ij}^2$	391	193	853	515	$\sum\sum x_{ij}^2 = 1952$

Step 4: Correction factor

$$\text{CF} = \frac{T^2}{N} = \frac{(-6)^2}{26} = \frac{36}{26} \approx 1.3846$$

Step 5: Sums of squares

$$\text{SST} = \sum\sum x_{ij}^2 - \text{CF} = 1952 - 1.3846 \approx 1950.6154$$

$$\begin{aligned}\text{SSC} &= \sum_j \frac{C_j^2}{n_j} - \text{CF} \\ &= \frac{29^2}{7} + \frac{11^2}{5} + \frac{(-3)^2}{8} + \frac{(-43)^2}{6} - \text{CF} \\ &\approx 120.1429 + 24.2 + 1.125 + 308.1667 - 1.3846 \\ &\approx 453.6345 - 1.3846 \approx 452.2499\end{aligned}$$

$$\text{SSE} = \text{SST} - \text{SSC} = 1950.6154 - 452.2499 \approx 1498.3655$$

(All sums of squares are in coded units; in hours^2 each is 100 times larger. The F-ratio is not affected.)

Step 6: Degrees of freedom and mean squares

$\nu_C = c - 1 = 3, \quad \nu_E = N - c = 26 - 4 = 22, \quad \nu_T = N - 1 = 25.$

$$\text{MSC} = \frac{\text{SSC}}{\nu_C} = \frac{452.2499}{3} \approx 150.7500$$

$$\text{MSE} = \frac{\text{SSE}}{\nu_E} = \frac{1498.3655}{22} \approx 68.1075$$

Step 7: F-ratio

$$F_C = \frac{\text{MSC}}{\text{MSE}} = \frac{150.7500}{68.1075} \approx 2.21$$

with $(\nu_C, \nu_E) = (3, 22)$ d.f.

Step 8: ANOVA table and decision

Source	SS	d.f.	MS	F_{cal}	$F_{0.05}(\nu, \nu_E)$	Decision
Columns (treatments)	452.2499	3	150.7500	2.21	3.05	Not significant
Error (within)	1498.3655	22	68.1075	–	–	–
Total	1950.6154	25				

- $F_C = 2.21 < F_{0.05}(3, 22) = 3.05 \Rightarrow H_0^{(C)}$ is accepted: the difference between the four brands is **not significant**.

Final Answer

The mean lives of the four brands are homogeneous; the brands do not differ significantly.

Two-Way ANOVA (Randomised Block Design)

Randomised Block Design (RBD)

A **Randomised Block Design** is used when experimental units differ systematically. Units are grouped into **blocks** according to a factor expected to influence the outcome, and the treatments are randomly assigned within each block. Grouping similar units gives more precise results and reduces the experimental error substantially. The data are analysed by **two-way ANOVA**.

Two-way layout

In a **two-way ANOVA** (one observation per cell, no replication) the data are classified according to two factors: one factor along the r **rows** (e.g. blocks) and the other along the c **columns** (e.g. treatments), with $N = rc$ observations. The method determines whether each factor has a significant effect and splits the total variation into the parts due to rows, columns and random error. (With one observation per cell the interaction cannot be separated from the error.)

Sums of squares

$$\text{CF} = \frac{T^2}{N}, \qquad \text{SST} = \sum_{i=1}^{r}\sum_{j=1}^{c} x_{ij}^2 - \text{CF},$$

$$\text{SSR} = \frac{1}{c}\sum_{i=1}^{r} R_i^2 - \text{CF} \quad \text{(between rows)},$$

$$\text{SSC} = \frac{1}{r}\sum_{j=1}^{c} C_j^2 - \text{CF} \quad \text{(between columns)},$$

$$\text{SSE} = \text{SST} - \text{SSR} - \text{SSC}.$$

Mean squares and F-ratios

$$\text{MSR} = \frac{\text{SSR}}{r-1}, \qquad \text{MSC} = \frac{\text{SSC}}{c-1}, \qquad \text{MSE} = \frac{\text{SSE}}{(r-1)(c-1)},$$

$$F_R = \frac{\text{MSR}}{\text{MSE}} \quad \text{with } (r-1,\ (r-1)(c-1)) \text{ d.f.},$$

$$F_C = \frac{\text{MSC}}{\text{MSE}} \quad \text{with } (c-1,\ (r-1)(c-1)) \text{ d.f.}$$

If F_R (or F_C) exceeds the corresponding table value, the row (or column) effect is significant.

Part II: Two-Way ANOVA – Worked Examples

Example 2.1: Crop Variety and Soil Type

Question

Two varieties of crops (A and B) are tested in three different soil types (1, 2 and 3). The yield (in kg) for each combination is recorded:

Soil type	Crop A	Crop B
1	10	12
2	9	11
3	13	14

Determine whether there is a significant difference in yield due to crop variety or soil type using two-way ANOVA (no replication) at 5% LOS.

Answer

Step 1: Hypotheses

- $H_0^{(R)}$: there is no significant difference between the soil types;
  $H_1^{(R)}$: there is a significant difference between the soil types.
- $H_0^{(C)}$: there is no significant difference between the crop varieties;
  $H_1^{(C)}$: there is a significant difference between the crop varieties.

Step 2: Data table with row and column totals

Soil type	Crop A	Crop B	R_i (row total)
1	10	12	22
2	9	11	20
3	13	14	27
C_j (column total)	32	37	$T = 69$

Check: $\sum R_i = \sum C_j = T = 69$, $r = 3$ rows, $c = 2$ columns, $N = rc = 6$.

Step 3: Table of squares x_{ij}^2
Squaring every observation:

Soil type	Crop A	Crop B	$\sum_j x_{ij}^2$
1	100	144	244
2	81	121	202
3	169	196	365
$\sum_i x_{ij}^2$	350	461	$\sum\sum x_{ij}^2 = 811$

Step 4: Correction factor

$$\text{CF} = \frac{T^2}{N} = \frac{69^2}{6} = \frac{4761}{6} = 793.5$$

Step 5: Sums of squares

Here $\text{SSR} = \frac{1}{c}\sum R_i^2 - \text{CF}$ and $\text{SSC} = \frac{1}{r}\sum C_j^2 - \text{CF}$.

$$\text{SST} = \sum\sum x_{ij}^2 - \text{CF} = 811 - 793.5 = 17.5$$

$$\begin{aligned}\text{SSR} &= \frac{22^2 + 20^2 + 27^2}{2} - \text{CF} \\ &= \frac{1613}{2} - 793.5 = 806.5 - 793.5 = 13\end{aligned}$$

$$\begin{aligned}\text{SSC} &= \frac{32^2 + 37^2}{3} - \text{CF} \\ &= \frac{2393}{3} - 793.5 \approx 797.6667 - 793.5 \approx 4.1667\end{aligned}$$

$$\begin{aligned}\text{SSE} &= \text{SST} - \text{SSR} - \text{SSC} \\ &= 17.5 - 13 - 4.1667 \approx 0.3333\end{aligned}$$

Step 6: Degrees of freedom and mean squares

$\nu_R = r - 1 = 2, \quad \nu_C = c - 1 = 1, \quad \nu_E = (r-1)(c-1) = 2, \quad \nu_T = N - 1 = 5.$

$$\text{MSR} = \frac{\text{SSR}}{\nu_R} = \frac{13}{2} = 6.5$$

$$\text{MSC} = \frac{\text{SSC}}{\nu_C} = \frac{4.1667}{1} \approx 4.1667$$

$$\text{MSE} = \frac{\text{SSE}}{\nu_E} = \frac{0.3333}{2} \approx 0.1667$$

Step 7: F-ratios

$$F_R = \frac{\text{MSR}}{\text{MSE}} = \frac{6.5}{0.1667} \approx 39.00$$

$$F_C = \frac{\text{MSC}}{\text{MSE}} = \frac{4.1667}{0.1667} \approx 25.00$$

F_R has $(\nu_R, \nu_E) = (2, 2)$ d.f.; F_C has $(\nu_C, \nu_E) = (1, 2)$ d.f.

Step 8: ANOVA table and decision

Source	SS	d.f.	MS	F_{cal}	$F_{0.05}(\nu, \nu_E)$	Decision
Rows	13	2	6.5	39.00	19.00	Significant
Columns	4.1667	1	4.1667	25.00	18.51	Significant
Error	0.3333	2	0.1667	–	–	–
Total	17.5	5				

- $F_R = 39.00 > F_{0.05}(2, 2) = 19.00 \Rightarrow H_0^{(R)}$ is rejected: the difference between the soil types is **significant**.
- $F_C = 25.00 > F_{0.05}(1, 2) = 18.51 \Rightarrow H_0^{(C)}$ is rejected: the difference between the crop varieties is **significant**.

Final Answer

Yield differs significantly with respect to both the soil type and the crop variety.

Example 2.2: Machines and Shifts

Question

A factory wants to test whether two machines (A and B) have different production times (in hours) during three shifts (Morning, Afternoon, Night). The data are:

Shift	Machine A	Machine B
Morning	8	10
Afternoon	9	8
Night	7	11

Perform a two-way ANOVA (shifts as rows, machines as columns) and comment on the significance of machine and shift effects at 5% LOS.

Answer

Step 1: Hypotheses

- $H_0^{(R)}$: there is no significant difference between the shifts;
  $H_1^{(R)}$: there is a significant difference between the shifts.
- $H_0^{(C)}$: there is no significant difference between the machines;
  $H_1^{(C)}$: there is a significant difference between the machines.

Step 2: Data table with row and column totals

Shift	Machine A	Machine B	R_i (row total)
Morning	8	10	18
Afternoon	9	8	17
Night	7	11	18
C_j (column total)	24	29	$T = 53$

Check: $\sum R_i = \sum C_j = T = 53$, $r = 3$ rows, $c = 2$ columns, $N = rc = 6$.

Step 3: Table of squares x_{ij}^2
Squaring every observation:

Shift	Machine A	Machine B	$\sum_j x_{ij}^2$
Morning	64	100	164
Afternoon	81	64	145
Night	49	121	170
$\sum_i x_{ij}^2$	194	285	$\sum\sum x_{ij}^2 = 479$

Step 4: Correction factor

$$\text{CF} = \frac{T^2}{N} = \frac{53^2}{6} = \frac{2809}{6} \approx 468.1667$$

Step 5: Sums of squares
Here $\text{SSR} = \frac{1}{c}\sum R_i^2 - \text{CF}$ and $\text{SSC} = \frac{1}{r}\sum C_j^2 - \text{CF}$.

$$\text{SST} = \sum\sum x_{ij}^2 - \text{CF} = 479 - 468.1667 \approx 10.8333$$

$$\begin{aligned}\text{SSR} &= \frac{18^2 + 17^2 + 18^2}{2} - \text{CF} \\ &= \frac{937}{2} - 468.1667 = 468.5 - 468.1667 \approx 0.3333\end{aligned}$$

$$\begin{aligned}\text{SSC} &= \frac{24^2 + 29^2}{3} - \text{CF} \\ &= \frac{1417}{3} - 468.1667 \approx 472.3333 - 468.1667 \approx 4.1667\end{aligned}$$

$$\begin{aligned}\text{SSE} &= \text{SST} - \text{SSR} - \text{SSC} \\ &= 10.8333 - 0.3333 - 4.1667 \approx 6.3333\end{aligned}$$

Step 6: Degrees of freedom and mean squares

$\nu_R = r-1 = 2, \quad \nu_C = c-1 = 1, \quad \nu_E = (r-1)(c-1) = 2, \quad \nu_T = N-1 = 5.$

$$\text{MSR} = \frac{\text{SSR}}{\nu_R} = \frac{0.3333}{2} \approx 0.1667$$
$$\text{MSC} = \frac{\text{SSC}}{\nu_C} = \frac{4.1667}{1} \approx 4.1667$$
$$\text{MSE} = \frac{\text{SSE}}{\nu_E} = \frac{6.3333}{2} \approx 3.1667$$

Step 7: F-ratios

$$F_R = \frac{\text{MSR}}{\text{MSE}} = \frac{0.1667}{3.1667} \approx 0.05$$
$$F_C = \frac{\text{MSC}}{\text{MSE}} = \frac{4.1667}{3.1667} \approx 1.32$$

F_R has $(\nu_R, \nu_E) = (2, 2)$ d.f.; F_C has $(\nu_C, \nu_E) = (1, 2)$ d.f.

Step 8: ANOVA table and decision

Source	SS	d.f.	MS	F_{cal}	$F_{0.05}(\nu, \nu_E)$	Decision
Rows	0.3333	2	0.1667	0.05	19.00	Not significant
Columns	4.1667	1	4.1667	1.32	18.51	Not significant
Error	6.3333	2	3.1667	–	–	–
Total	10.8333	5				

- $F_R = 0.05 < F_{0.05}(2,2) = 19.00$ (in fact $F < 1$) $\Rightarrow H_0^{(R)}$ is accepted: the difference between the shifts is **not significant**.
- $F_C = 1.32 < F_{0.05}(1,2) = 18.51 \Rightarrow H_0^{(C)}$ is accepted: the difference between the machines is **not significant**.

Final Answer

Neither the shifts nor the machines have a significant effect on production time.

Example 2.3: Randomised Block Design (Three Crops)

Question

Three varieties of a crop are tested in a randomised block design with four replications (blocks). The yields (in kg) are:

Block	A	B	C
1	47	49	48
2	51	49	53
3	49	52	52
4	49	50	51

Analyse the data for significance of crop varieties and block effects at 5% LOS.

Answer

Step 1: Hypotheses

- $H_0^{(R)}$: there is no significant difference between the blocks;
  $H_1^{(R)}$: there is a significant difference between the blocks.
- $H_0^{(C)}$: there is no significant difference between the crop varieties;
  $H_1^{(C)}$: there is a significant difference between the crop varieties.

Step 2: Data table with row and column totals

Block	A	B	C	R_i (row total)
1	47	49	48	144
2	51	49	53	153
3	49	52	52	153
4	49	50	51	150
C_j (column total)	196	200	204	$T = 600$

Check: $\sum R_i = \sum C_j = T = 600$, $r = 4$ rows, $c = 3$ columns, $N = rc = 12$.

Step 3: Table of squares x_{ij}^2
Squaring every observation:

Block	A	B	C	$\sum_j x_{ij}^2$
1	2209	2401	2304	6914
2	2601	2401	2809	7811
3	2401	2704	2704	7809
4	2401	2500	2601	7502
$\sum_i x_{ij}^2$	9612	10006	10418	$\sum\sum x_{ij}^2 = 30036$

Step 4: Correction factor

$$\text{CF} = \frac{T^2}{N} = \frac{600^2}{12} = \frac{360000}{12} = 30000$$

Step 5: Sums of squares

Here $\text{SSR} = \frac{1}{c}\sum R_i^2 - \text{CF}$ and $\text{SSC} = \frac{1}{r}\sum C_j^2 - \text{CF}$.

$$\text{SST} = \sum\sum x_{ij}^2 - \text{CF} = 30036 - 30000 = 36$$

$$\begin{aligned}\text{SSR} &= \frac{144^2 + 153^2 + 153^2 + 150^2}{3} - \text{CF} \\ &= \frac{90054}{3} - 30000 = 30018 - 30000 = 18\end{aligned}$$

$$\begin{aligned}\text{SSC} &= \frac{196^2 + 200^2 + 204^2}{4} - \text{CF} \\ &= \frac{120032}{4} - 30000 = 30008 - 30000 = 8\end{aligned}$$

$$\begin{aligned}\text{SSE} &= \text{SST} - \text{SSR} - \text{SSC} \\ &= 36 - 18 - 8 = 10\end{aligned}$$

Step 6: Degrees of freedom and mean squares

$\nu_R = r - 1 = 3, \quad \nu_C = c - 1 = 2, \quad \nu_E = (r-1)(c-1) = 6, \quad \nu_T = N - 1 = 11.$

$$\text{MSR} = \frac{\text{SSR}}{\nu_R} = \frac{18}{3} = 6$$

$$\text{MSC} = \frac{\text{SSC}}{\nu_C} = \frac{8}{2} = 4$$

$$\text{MSE} = \frac{\text{SSE}}{\nu_E} = \frac{10}{6} \approx 1.6667$$

Step 7: F-ratios

$$F_R = \frac{\text{MSR}}{\text{MSE}} = \frac{6}{1.6667} \approx 3.60$$

$$F_C = \frac{\text{MSC}}{\text{MSE}} = \frac{4}{1.6667} \approx 2.40$$

F_R has $(\nu_R, \nu_E) = (3, 6)$ d.f.; F_C has $(\nu_C, \nu_E) = (2, 6)$ d.f.

Step 8: ANOVA table and decision

Source	SS	d.f.	MS	F_{cal}	$F_{0.05}(\nu, \nu_E)$	Decision
Rows	18	3	6	3.60	4.76	Not significant
Columns	8	2	4	2.40	5.14	Not significant
Error	10	6	1.6667	–	–	–
Total	36	11				

- $F_R = 3.60 < F_{0.05}(3, 6) = 4.76 \Rightarrow H_0^{(R)}$ is accepted: the difference between the blocks is **not significant**.
- $F_C = 2.40 < F_{0.05}(2, 6) = 5.14 \Rightarrow H_0^{(C)}$ is accepted: the difference between the crop varieties is **not significant**.

Final Answer

There is no significant difference between the three crop varieties, nor a significant block effect.

Latin Square Design (Three-Way ANOVA)

Latin Square Design (LSD)

A **Latin Square Design** of order n arranges n treatments in an $n \times n$ square so that each treatment (letter) occurs exactly once in every row and once in every column. It is used in agricultural, industrial or laboratory experiments when two extraneous sources of variability must be controlled; it is efficient with limited resources.

- **Rows:** first blocking factor (e.g. plots, days).
- **Columns:** second blocking factor (e.g. soil types, engines).
- **Treatments (letters):** the factor under study; totals T_k.

Here $N = n^2$ and $T = \sum R_i = \sum C_j = \sum T_k$.

Sums of squares

$$\text{CF} = \frac{T^2}{N}, \qquad \text{SST} = \sum_{i=1}^{n}\sum_{j=1}^{n} x_{ij}^2 - \text{CF},$$

$$\text{SSR} = \frac{1}{n}\sum_{i=1}^{n} R_i^2 - \text{CF}, \qquad \text{SSC} = \frac{1}{n}\sum_{j=1}^{n} C_j^2 - \text{CF}, \qquad \text{SSTr} = \frac{1}{n}\sum_{k=1}^{n} T_k^2 - \text{CF},$$

$$\text{SSE} = \text{SST} - \text{SSR} - \text{SSC} - \text{SSTr}.$$

Mean squares and F-ratios

$$\text{MSR} = \frac{\text{SSR}}{n-1}, \quad \text{MSC} = \frac{\text{SSC}}{n-1}, \quad \text{MSTr} = \frac{\text{SSTr}}{n-1}, \quad \text{MSE} = \frac{\text{SSE}}{(n-1)(n-2)},$$

$$F_R = \frac{\text{MSR}}{\text{MSE}}, \qquad F_C = \frac{\text{MSC}}{\text{MSE}}, \qquad F_{Tr} = \frac{\text{MSTr}}{\text{MSE}},$$

each with $(n-1,\ (n-1)(n-2))$ d.f. If an F-ratio exceeds the table value, the corresponding effect is significant.

Part III: Latin Square Design – Worked Examples

Example 3.1: Yield of Crops under Different Fertilizers

Question

Three types of fertilizers (A, B, C) are tested on crop yield, with three plots (rows) and three soil types (columns), in a 3×3 Latin square:

	Soil 1	Soil 2	Soil 3
Plot 1	A: 6	B: 7	C: 8
Plot 2	B: 8	C: 9	A: 7
Plot 3	C: 5	A: 6	B: 8

Test whether there are significant differences between plots, soil types and fertilizers at 5% LOS.

Answer

Step 1: Hypotheses

- $H_0^{(R)}$: there is no significant difference between the plots;
  $H_1^{(R)}$: there is a significant difference between the plots.
- $H_0^{(C)}$: there is no significant difference between the soil types;
  $H_1^{(C)}$: there is a significant difference between the soil types.
- $H_0^{(Tr)}$: there is no significant difference between the fertilizers;
  $H_1^{(Tr)}$: there is a significant difference between the fertilizers.

Step 2: Data table with row, column and treatment totals

Plot	Soil 1	Soil 2	Soil 3	R_i (row total)
Plot 1	A: 6	B: 7	C: 8	21
Plot 2	B: 8	C: 9	A: 7	24
Plot 3	C: 5	A: 6	B: 8	19
C_j (column total)	19	22	23	$T = 64$

Check: $\sum R_i = \sum C_j = T = 64$, order $n = 3$, $N = n^2 = 9$.
Collecting the observations of each treatment (letter):

Treatment	Observations	T_k (treatment total)
A	6 7 6	$T_A = 19$
B	7 8 8	$T_B = 23$
C	8 9 5	$T_C = 22$
	Check: $\sum T_k$	$T = 64$

Step 3: Table of squares x_{ij}^2
Squaring every observation:

Plot	Soil 1	Soil 2	Soil 3	$\sum_j x_{ij}^2$
Plot 1	36	49	64	149
Plot 2	64	81	49	194
Plot 3	25	36	64	125
$\sum_i x_{ij}^2$	125	166	177	$\sum\sum x_{ij}^2 = 468$

Step 4: Correction factor

$$\text{CF} = \frac{T^2}{N} = \frac{64^2}{9} = \frac{4096}{9} \approx 455.1111$$

Step 5: Sums of squares

Here $\text{SSR} = \frac{1}{n}\sum R_i^2 - \text{CF}$, $\text{SSC} = \frac{1}{n}\sum C_j^2 - \text{CF}$, $\text{SSTr} = \frac{1}{n}\sum T_k^2 - \text{CF}$.

$$\text{SST} = \sum\sum x_{ij}^2 - \text{CF} = 468 - 455.1111 \approx 12.8889$$

$$\begin{aligned}\text{SSR} &= \frac{21^2 + 24^2 + 19^2}{3} - \text{CF} \\ &= \frac{1378}{3} - 455.1111 \approx 459.3333 - 455.1111 \approx 4.2222\end{aligned}$$

$$\begin{aligned}\text{SSC} &= \frac{19^2 + 22^2 + 23^2}{3} - \text{CF} \\ &= \frac{1374}{3} - 455.1111 = 458 - 455.1111 \approx 2.8889\end{aligned}$$

$$\begin{aligned}\text{SSTr} &= \frac{19^2 + 23^2 + 22^2}{3} - \text{CF} \\ &= \frac{1374}{3} - 455.1111 = 458 - 455.1111 \approx 2.8889\end{aligned}$$

$$\begin{aligned}\text{SSE} &= \text{SST} - \text{SSR} - \text{SSC} - \text{SSTr} \\ &= 12.8889 - 4.2222 - 2.8889 - 2.8889 \approx 2.8889\end{aligned}$$

Step 6: Degrees of freedom and mean squares

$\nu_R = \nu_C = \nu_{Tr} = n - 1 = 2, \quad \nu_E = (n-1)(n-2) = 2, \quad \nu_T = n^2 - 1 = 8.$

$$\text{MSR} = \frac{\text{SSR}}{\nu_R} = \frac{4.2222}{2} \approx 2.1111$$

$$\text{MSC} = \frac{\text{SSC}}{\nu_C} = \frac{2.8889}{2} \approx 1.4444$$

$$\text{MSTr} = \frac{\text{SSTr}}{\nu_{Tr}} = \frac{2.8889}{2} \approx 1.4444$$

$$\text{MSE} = \frac{\text{SSE}}{\nu_E} = \frac{2.8889}{2} \approx 1.4444$$

Step 7: F-ratios

$$F_R = \frac{\text{MSR}}{\text{MSE}} = \frac{2.1111}{1.4444} \approx 1.46$$
$$F_C = \frac{\text{MSC}}{\text{MSE}} = \frac{1.4444}{1.4444} \approx 1.00$$
$$F_{Tr} = \frac{\text{MSTr}}{\text{MSE}} = \frac{1.4444}{1.4444} \approx 1.00$$

All three ratios have $(n-1, (n-1)(n-2)) = (2, 2)$ d.f.

Step 8: ANOVA table and decision

Source	SS	d.f.	MS	F_{cal}	$F_{0.05}(\nu, \nu_E)$	Decision
Rows	4.2222	2	2.1111	1.46	19.00	Not significant
Columns	2.8889	2	1.4444	1.00	19.00	Not significant
Treatments	2.8889	2	1.4444	1.00	19.00	Not significant
Error	2.8889	2	1.4444	–	–	–
Total	12.8889	8				

- $F_R = 1.46 < F_{0.05}(2, 2) = 19.00 \Rightarrow H_0^{(R)}$ is accepted: the difference between the plots is **not significant**.
- $F_C = 1.00 < F_{0.05}(2, 2) = 19.00 \Rightarrow H_0^{(C)}$ is accepted: the difference between the soil types is **not significant**.
- $F_{Tr} = 1.00 < F_{0.05}(2, 2) = 19.00 \Rightarrow H_0^{(Tr)}$ is accepted: the difference between the fertilizers is **not significant**.

Final Answer

There is no significant difference in yield between plots, soil types or fertilizers.

Example 3.2: Comparison of Burners

Question

The following data resulted from an experiment to compare three burners B_1, B_2 and B_3. A Latin square design was used as the tests were made on three engines and were spread over three days.

	Engine 1	Engine 2	Engine 3
Day 1	B_1: 16	B_2: 17	B_3: 20
Day 2	B_2: 16	B_3: 21	B_1: 15
Day 3	B_3: 15	B_1: 12	B_2: 13

Test the hypothesis that there is no difference between the burners at 5% LOS.

Answer

Step 1: Hypotheses

- $H_0^{(R)}$: there is no significant difference between the days;
  $H_1^{(R)}$: there is a significant difference between the days.
- $H_0^{(C)}$: there is no significant difference between the engines;
  $H_1^{(C)}$: there is a significant difference between the engines.
- $H_0^{(Tr)}$: there is no significant difference between the burners;
  $H_1^{(Tr)}$: there is a significant difference between the burners.

Step 2: Data table with row, column and treatment totals

Day	Engine 1	Engine 2	Engine 3	R_i (row total)
Day 1	B_1: 16	B_2: 17	B_3: 20	53
Day 2	B_2: 16	B_3: 21	B_1: 15	52
Day 3	B_3: 15	B_1: 12	B_2: 13	40
C_j (column total)	47	50	48	$T = 145$

Check: $\sum R_i = \sum C_j = T = 145$, order $n = 3$, $N = n^2 = 9$.
Collecting the observations of each treatment (letter):

Treatment	Observations	T_k (treatment total)
B_1	16 15 12	$T_{B_1} = 43$
B_2	17 16 13	$T_{B_2} = 46$
B_3	20 21 15	$T_{B_3} = 56$
	Check: $\sum T_k$	$T = 145$

Step 3: Table of squares x_{ij}^2
Squaring every observation:

Day	Engine 1	Engine 2	Engine 3	$\sum_j x_{ij}^2$
Day 1	256	289	400	945
Day 2	256	441	225	922
Day 3	225	144	169	538
$\sum_i x_{ij}^2$	737	874	794	$\sum\sum x_{ij}^2 = 2405$

Step 4: Correction factor

$$\text{CF} = \frac{T^2}{N} = \frac{145^2}{9} = \frac{21025}{9} \approx 2336.1111$$

Step 5: Sums of squares

Here $\text{SSR} = \frac{1}{n}\sum R_i^2 - \text{CF}$, $\text{SSC} = \frac{1}{n}\sum C_j^2 - \text{CF}$, $\text{SSTr} = \frac{1}{n}\sum T_k^2 - \text{CF}$.

$$\text{SST} = \sum\sum x_{ij}^2 - \text{CF} = 2405 - 2336.1111 \approx 68.8889$$

$$\begin{aligned}\text{SSR} &= \frac{53^2 + 52^2 + 40^2}{3} - \text{CF} \\ &= \frac{7113}{3} - 2336.1111 = 2371 - 2336.1111 \approx 34.8889\end{aligned}$$

$$\begin{aligned}\text{SSC} &= \frac{47^2 + 50^2 + 48^2}{3} - \text{CF} \\ &= \frac{7013}{3} - 2336.1111 \approx 2337.6667 - 2336.1111 \approx 1.5556\end{aligned}$$

$$\begin{aligned}\text{SSTr} &= \frac{43^2 + 46^2 + 56^2}{3} - \text{CF} \\ &= \frac{7101}{3} - 2336.1111 = 2367 - 2336.1111 \approx 30.8889\end{aligned}$$

$$\begin{aligned}\text{SSE} &= \text{SST} - \text{SSR} - \text{SSC} - \text{SSTr} \\ &= 68.8889 - 34.8889 - 1.5556 - 30.8889 \approx 1.5556\end{aligned}$$

Step 6: Degrees of freedom and mean squares

$\nu_R = \nu_C = \nu_{Tr} = n - 1 = 2, \quad \nu_E = (n-1)(n-2) = 2, \quad \nu_T = n^2 - 1 = 8.$

$$\text{MSR} = \frac{\text{SSR}}{\nu_R} = \frac{34.8889}{2} \approx 17.4444$$

$$\text{MSC} = \frac{\text{SSC}}{\nu_C} = \frac{1.5556}{2} \approx 0.7778$$

$$\text{MSTr} = \frac{\text{SSTr}}{\nu_{Tr}} = \frac{30.8889}{2} \approx 15.4444$$

$$\text{MSE} = \frac{\text{SSE}}{\nu_E} = \frac{1.5556}{2} \approx 0.7778$$

Step 7: F-ratios

$$F_R = \frac{\text{MSR}}{\text{MSE}} = \frac{17.4444}{0.7778} \approx 22.43$$

$$F_C = \frac{\text{MSC}}{\text{MSE}} = \frac{0.7778}{0.7778} \approx 1.00$$

$$F_{Tr} = \frac{\text{MSTr}}{\text{MSE}} = \frac{15.4444}{0.7778} \approx 19.86$$

All three ratios have $(n-1, (n-1)(n-2)) = (2, 2)$ d.f.

Step 8: ANOVA table and decision

Source	SS	d.f.	MS	F_{cal}	$F_{0.05}(\nu, \nu_E)$	Decision
Rows	34.8889	2	17.4444	22.43	19.00	Significant
Columns	1.5556	2	0.7778	1.00	19.00	Not significant
Treatments	30.8889	2	15.4444	19.86	19.00	Significant
Error	1.5556	2	0.7778	–	–	–
Total	68.8889	8				

- $F_R = 22.43 > F_{0.05}(2,2) = 19.00 \Rightarrow H_0^{(R)}$ is rejected: the difference between the days is **significant**.
- $F_C = 1.00 < F_{0.05}(2,2) = 19.00 \Rightarrow H_0^{(C)}$ is accepted: the difference between the engines is **not significant**.
- $F_{Tr} = 19.86 > F_{0.05}(2,2) = 19.00 \Rightarrow H_0^{(Tr)}$ is rejected: the difference between the burners is **significant**.

Final Answer

The burners differ significantly (so do the days); the engine differences are not significant. Hence the three burners do not have the same mean performance.

Example 3.3: Methods of Cultivation

Question

Analyse the variance in the following Latin square of yields (in kg) of paddy, where A, B, C and D denote four different methods of cultivation.

	Col 1	Col 2	Col 3	Col 4
Row 1	D: 122	A: 121	C: 123	B: 122
Row 2	B: 124	C: 123	A: 122	D: 125
Row 3	A: 120	B: 119	D: 120	C: 121
Row 4	C: 122	D: 123	B: 121	A: 122

Examine whether the different methods of cultivation have given significantly different yields at 5% LOS.

Answer

Step 1: Hypotheses

- $H_0^{(R)}$: there is no significant difference between the rows;
  $H_1^{(R)}$: there is a significant difference between the rows.
- $H_0^{(C)}$: there is no significant difference between the columns;
  $H_1^{(C)}$: there is a significant difference between the columns.
- $H_0^{(Tr)}$: there is no significant difference between the methods of cultivation;
  $H_1^{(Tr)}$: there is a significant difference between the methods of cultivation.

Step 2: Data table with row, column and treatment totals

Row	Col 1	Col 2	Col 3	Col 4	R_i (row total)
Row 1	D: 122	A: 121	C: 123	B: 122	488
Row 2	B: 124	C: 123	A: 122	D: 125	494
Row 3	A: 120	B: 119	D: 120	C: 121	480
Row 4	C: 122	D: 123	B: 121	A: 122	488
C_j (column total)	488	486	486	490	$T = 1950$

Check: $\sum R_i = \sum C_j = T = 1950$, order $n = 4$, $N = n^2 = 16$.
Collecting the observations of each treatment (letter):

Treatment	Observations				T_k (treatment total)
A	121	122	120	122	$T_A = 485$
B	122	124	119	121	$T_B = 486$
C	123	123	121	122	$T_C = 489$
D	122	125	120	123	$T_D = 490$
			Check: $\sum T_k$		$T = 1950$

Step 3: Table of squares x_{ij}^2
Squaring every observation:

Row	Col 1	Col 2	Col 3	Col 4	$\sum_j x_{ij}^2$
Row 1	14884	14641	15129	14884	59538
Row 2	15376	15129	14884	15625	61014
Row 3	14400	14161	14400	14641	57602
Row 4	14884	15129	14641	14884	59538
$\sum_i x_{ij}^2$	59544	59060	59054	60034	$\sum\sum x_{ij}^2 = 237692$

Step 4: Correction factor

$$\text{CF} = \frac{T^2}{N} = \frac{1950^2}{16} = \frac{3802500}{16} = 237656.25$$

Step 5: Sums of squares
Here $\text{SSR} = \frac{1}{n}\sum R_i^2 - \text{CF}$, $\text{SSC} = \frac{1}{n}\sum C_j^2 - \text{CF}$, $\text{SSTr} = \frac{1}{n}\sum T_k^2 - \text{CF}$.

$$\text{SST} = \sum\sum x_{ij}^2 - \text{CF} = 237692 - 237656.25 = 35.75$$

$$\begin{aligned}\text{SSR} &= \frac{488^2 + 494^2 + 480^2 + 488^2}{4} - \text{CF} \\ &= \frac{950724}{4} - 237656.25 = 237681 - 237656.25 = 24.75\end{aligned}$$

$$\begin{aligned}\text{SSC} &= \frac{488^2 + 486^2 + 486^2 + 490^2}{4} - \text{CF} \\ &= \frac{950636}{4} - 237656.25 = 237659 - 237656.25 = 2.75\end{aligned}$$

$$\begin{aligned}\text{SSTr} &= \frac{485^2 + 486^2 + 489^2 + 490^2}{4} - \text{CF} \\ &= \frac{950642}{4} - 237656.25 = 237660.5 - 237656.25 = 4.25\end{aligned}$$

$$\begin{aligned}\text{SSE} &= \text{SST} - \text{SSR} - \text{SSC} - \text{SSTr} \\ &= 35.75 - 24.75 - 2.75 - 4.25 = 4\end{aligned}$$

Step 6: Degrees of freedom and mean squares

$\nu_R = \nu_C = \nu_{Tr} = n - 1 = 3, \quad \nu_E = (n-1)(n-2) = 6, \quad \nu_T = n^2 - 1 = 15.$

$$\text{MSR} = \frac{\text{SSR}}{\nu_R} = \frac{24.75}{3} = 8.25$$
$$\text{MSC} = \frac{\text{SSC}}{\nu_C} = \frac{2.75}{3} \approx 0.9167$$
$$\text{MSTr} = \frac{\text{SSTr}}{\nu_{Tr}} = \frac{4.25}{3} \approx 1.4167$$
$$\text{MSE} = \frac{\text{SSE}}{\nu_E} = \frac{4}{6} \approx 0.6667$$

Step 7: F-ratios

$$F_R = \frac{\text{MSR}}{\text{MSE}} = \frac{8.25}{0.6667} \approx 12.38$$
$$F_C = \frac{\text{MSC}}{\text{MSE}} = \frac{0.9167}{0.6667} \approx 1.38$$
$$F_{Tr} = \frac{\text{MSTr}}{\text{MSE}} = \frac{1.4167}{0.6667} \approx 2.12$$

All three ratios have $(n-1, (n-1)(n-2)) = (3, 6)$ d.f.

Step 8: ANOVA table and decision

Source	SS	d.f.	MS	F_{cal}	$F_{0.05}(\nu, \nu_E)$	Decision
Rows	24.75	3	8.25	12.38	4.76	Significant
Columns	2.75	3	0.9167	1.38	4.76	Not significant
Treatments	4.25	3	1.4167	2.12	4.76	Not significant
Error	4	6	0.6667	–	–	–
Total	35.75	15				

- $F_R = 12.38 > F_{0.05}(3,6) = 4.76 \Rightarrow H_0^{(R)}$ is rejected: the difference between the rows is **significant**.
- $F_C = 1.38 < F_{0.05}(3,6) = 4.76 \Rightarrow H_0^{(C)}$ is accepted: the difference between the columns is **not significant**.
- $F_{Tr} = 2.12 < F_{0.05}(3,6) = 4.76 \Rightarrow H_0^{(Tr)}$ is accepted: the difference between the methods of cultivation is **not significant**.

Final Answer

The rows differ significantly, but the methods of cultivation (and the columns) do not give significantly different yields.

ANOVA Question Bank (with Solutions)

Question 1: Doctors and Treatments

Question

Four doctors each test four treatments for a certain disease and observe the number of days each patient takes to recover. The results (recovery time in days) are:

Doctor	Treatment 1	Treatment 2	Treatment 3	Treatment 4
A	10	14	19	20
B	11	15	17	21
C	9	12	16	19
D	8	13	17	20

Discuss the difference between (a) doctors and (b) treatments at 5% LOS.

Answer

Step 1: Hypotheses

- $H_0^{(R)}$: there is no significant difference between the doctors;
  $H_1^{(R)}$: there is a significant difference between the doctors.
- $H_0^{(C)}$: there is no significant difference between the treatments;
  $H_1^{(C)}$: there is a significant difference between the treatments.

Step 2: Data table with row and column totals

Doctor	T1	T2	T3	T4	R_i (row total)
A	10	14	19	20	63
B	11	15	17	21	64
C	9	12	16	19	56
D	8	13	17	20	58
C_j (column total)	38	54	69	80	$T = 241$

Check: $\sum R_i = \sum C_j = T = 241$, $r = 4$ rows, $c = 4$ columns, $N = rc = 16$.

Step 3: Table of squares x_{ij}^2

Squaring every observation:

Doctor	T1	T2	T3	T4	$\sum_j x_{ij}^2$
A	100	196	361	400	1057
B	121	225	289	441	1076
C	81	144	256	361	842
D	64	169	289	400	922
$\sum_i x_{ij}^2$	366	734	1195	1602	$\sum\sum x_{ij}^2 = 3897$

Step 4: Correction factor

$$\text{CF} = \frac{T^2}{N} = \frac{241^2}{16} = \frac{58081}{16} = 3630.0625$$

Step 5: Sums of squares

Here $\text{SSR} = \frac{1}{c}\sum R_i^2 - \text{CF}$ and $\text{SSC} = \frac{1}{r}\sum C_j^2 - \text{CF}$.

$$\text{SST} = \sum\sum x_{ij}^2 - \text{CF} = 3897 - 3630.0625 = 266.9375$$

$$\begin{aligned}\text{SSR} &= \frac{63^2 + 64^2 + 56^2 + 58^2}{4} - \text{CF} \\ &= \frac{14565}{4} - 3630.0625 = 3641.25 - 3630.0625 = 11.1875\end{aligned}$$

$$\begin{aligned}\text{SSC} &= \frac{38^2 + 54^2 + 69^2 + 80^2}{4} - \text{CF} \\ &= \frac{15521}{4} - 3630.0625 = 3880.25 - 3630.0625 = 250.1875\end{aligned}$$

$$\begin{aligned}\text{SSE} &= \text{SST} - \text{SSR} - \text{SSC} \\ &= 266.9375 - 11.1875 - 250.1875 = 5.5625\end{aligned}$$

Step 6: Degrees of freedom and mean squares

$\nu_R = r - 1 = 3, \quad \nu_C = c - 1 = 3, \quad \nu_E = (r-1)(c-1) = 9, \quad \nu_T = N - 1 = 15.$

$$\text{MSR} = \frac{\text{SSR}}{\nu_R} = \frac{11.1875}{3} \approx 3.7292$$
$$\text{MSC} = \frac{\text{SSC}}{\nu_C} = \frac{250.1875}{3} \approx 83.3958$$
$$\text{MSE} = \frac{\text{SSE}}{\nu_E} = \frac{5.5625}{9} \approx 0.6181$$

Step 7: F-ratios

$$F_R = \frac{\text{MSR}}{\text{MSE}} = \frac{3.7292}{0.6181} \approx 6.03$$
$$F_C = \frac{\text{MSC}}{\text{MSE}} = \frac{83.3958}{0.6181} \approx 134.93$$

F_R has $(\nu_R, \nu_E) = (3, 9)$ d.f.; F_C has $(\nu_C, \nu_E) = (3, 9)$ d.f.

Step 8: ANOVA table and decision

Source	SS	d.f.	MS	F_{cal}	$F_{0.05}(\nu, \nu_E)$	Decision
Rows	11.1875	3	3.7292	6.03	3.86	Significant
Columns	250.1875	3	83.3958	134.93	3.86	Significant
Error	5.5625	9	0.6181	–	–	–
Total	266.9375	15				

- $F_R = 6.03 > F_{0.05}(3, 9) = 3.86 \Rightarrow H_0^{(R)}$ is rejected: the difference between the doctors is **significant**.
- $F_C = 134.93 > F_{0.05}(3, 9) = 3.86 \Rightarrow H_0^{(C)}$ is rejected: the difference between the treatments is **significant**.

Final Answer

(a) The doctors differ significantly. (b) The treatments differ significantly.

Question 2: Workers and Machines

Question

The following data represent the number of units of production per day turned out by five different workers using four different types of machines:

Worker	Machine A	Machine B	Machine C	Machine D
1	44	38	47	36
2	46	40	52	43
3	34	36	44	32
4	43	38	46	33
5	38	42	49	39

(a) Test whether the five workers differ with respect to mean productivity.
(b) Test whether the mean productivity is the same for the four different machine types.

Answer

Step 1: Hypotheses

- $H_0^{(R)}$: there is no significant difference between the workers;
  $H_1^{(R)}$: there is a significant difference between the workers.
- $H_0^{(C)}$: there is no significant difference between the machine types;
  $H_1^{(C)}$: there is a significant difference between the machine types.

Step 2: Data table with row and column totals

Worker	A	B	C	D	R_i (row total)
1	44	38	47	36	165
2	46	40	52	43	181
3	34	36	44	32	146
4	43	38	46	33	160
5	38	42	49	39	168
C_j (column total)	205	194	238	183	$T = 820$

Check: $\sum R_i = \sum C_j = T = 820$, $r = 5$ rows, $c = 4$ columns, $N = rc = 20$.

Step 3: Table of squares x_{ij}^2

Squaring every observation:

Worker	A	B	C	D	$\sum_j x_{ij}^2$
1	1936	1444	2209	1296	6885
2	2116	1600	2704	1849	8269
3	1156	1296	1936	1024	5412
4	1849	1444	2116	1089	6498
5	1444	1764	2401	1521	7130
$\sum_i x_{ij}^2$	8501	7548	11366	6779	$\sum\sum x_{ij}^2 = 34194$

Step 4: Correction factor

$$\text{CF} = \frac{T^2}{N} = \frac{820^2}{20} = \frac{672400}{20} = 33620$$

Step 5: Sums of squares

Here $\text{SSR} = \frac{1}{c}\sum R_i^2 - \text{CF}$ and $\text{SSC} = \frac{1}{r}\sum C_j^2 - \text{CF}$.

$$\text{SST} = \sum\sum x_{ij}^2 - \text{CF} = 34194 - 33620 = 574$$

$$\begin{aligned}\text{SSR} &= \frac{165^2 + 181^2 + 146^2 + 160^2 + 168^2}{4} - \text{CF} \\ &= \frac{135126}{4} - 33620 = 33781.5 - 33620 = 161.5\end{aligned}$$

$$\begin{aligned}\text{SSC} &= \frac{205^2 + 194^2 + 238^2 + 183^2}{5} - \text{CF} \\ &= \frac{169794}{5} - 33620 = 33958.8 - 33620 = 338.8\end{aligned}$$

$$\begin{aligned}\text{SSE} &= \text{SST} - \text{SSR} - \text{SSC} \\ &= 574 - 161.5 - 338.8 = 73.7\end{aligned}$$

Step 6: Degrees of freedom and mean squares

$\nu_R = r - 1 = 4, \quad \nu_C = c - 1 = 3, \quad \nu_E = (r-1)(c-1) = 12, \quad \nu_T = N - 1 = 19.$

$$\text{MSR} = \frac{\text{SSR}}{\nu_R} = \frac{161.5}{4} = 40.375$$
$$\text{MSC} = \frac{\text{SSC}}{\nu_C} = \frac{338.8}{3} \approx 112.9333$$
$$\text{MSE} = \frac{\text{SSE}}{\nu_E} = \frac{73.7}{12} \approx 6.1417$$

Step 7: F-ratios

$$F_R = \frac{\text{MSR}}{\text{MSE}} = \frac{40.375}{6.1417} \approx 6.57$$
$$F_C = \frac{\text{MSC}}{\text{MSE}} = \frac{112.9333}{6.1417} \approx 18.39$$

F_R has $(\nu_R, \nu_E) = (4, 12)$ d.f.; F_C has $(\nu_C, \nu_E) = (3, 12)$ d.f.

Step 8: ANOVA table and decision

Source	SS	d.f.	MS	F_{cal}	$F_{0.05}(\nu, \nu_E)$	Decision
Rows	161.5	4	40.375	6.57	3.26	Significant
Columns	338.8	3	112.9333	18.39	3.49	Significant
Error	73.7	12	6.1417	–	–	–
Total	574	19				

- $F_R = 6.57 > F_{0.05}(4, 12) = 3.26 \Rightarrow H_0^{(R)}$ is rejected: the difference between the workers is **significant**.
- $F_C = 18.39 > F_{0.05}(3, 12) = 3.49 \Rightarrow H_0^{(C)}$ is rejected: the difference between the machine types is **significant**.

Final Answer

(a) The five workers differ significantly in mean productivity. (b) The mean productivity is not the same for the four machine types.

Question 3: Durability of Computers

Question

In order to determine whether there is a significant difference in the durability of three makes of computers, samples of size five are selected from each make and the frequency of repairs during the first year of purchase is observed. The results are:

Make A	Make B	Make C
5	8	7
6	10	3
8	11	5
9	12	4
7	4	1

State the conclusion based on ANOVA at 5% LOS.

Answer

Step 1: Hypotheses

- $H_0^{(C)}$: there is no significant difference between the three makes;
  $H_1^{(C)}$: there is a significant difference between the three makes.

Step 2: Data table with row and column totals

Treatments are placed column-wise. Column totals C_j give the treatment totals; row totals R_i are used only as a check.

Obs.	Make A	Make B	Make C	R_i (row total)
1	5	8	7	20
2	6	10	3	19
3	8	11	5	24
4	9	12	4	25
5	7	4	1	12
C_j (column total)	35	45	20	$T = 100$
n_j (no. of obs.)	5	5	5	$N = 15$

Check: $\sum R_i = \sum C_j = T = 100$; $c = 3$, $N = 15$.

Step 3: Table of squares x_{ij}^2

Squaring every observation:

Obs.	Make A	Make B	Make C	$\sum_j x_{ij}^2$
1	25	64	49	138
2	36	100	9	145
3	64	121	25	210
4	81	144	16	241
5	49	16	1	66
$\sum_i x_{ij}^2$	255	445	100	$\sum\sum x_{ij}^2 = 800$

Step 4: Correction factor

$$\text{CF} = \frac{T^2}{N} = \frac{100^2}{15} = \frac{10000}{15} \approx 666.6667$$

Step 5: Sums of squares

$$\text{SST} = \sum\sum x_{ij}^2 - \text{CF} = 800 - 666.6667 \approx 133.3333$$

$$\begin{aligned}\text{SSC} &= \sum_j \frac{C_j^2}{n_j} - \text{CF} \\ &= \frac{35^2}{5} + \frac{45^2}{5} + \frac{20^2}{5} - \text{CF} \\ &= 245 + 405 + 80 - 666.6667 \\ &= 730 - 666.6667 \approx 63.3333\end{aligned}$$

$$\text{SSE} = \text{SST} - \text{SSC} = 133.3333 - 63.3333 = 70$$

Step 6: Degrees of freedom and mean squares

$\nu_C = c - 1 = 2, \quad \nu_E = N - c = 15 - 3 = 12, \quad \nu_T = N - 1 = 14.$

$$\text{MSC} = \frac{\text{SSC}}{\nu_C} = \frac{63.3333}{2} \approx 31.6667$$

$$\text{MSE} = \frac{\text{SSE}}{\nu_E} = \frac{70}{12} \approx 5.8333$$

Step 7: F-ratio

$$F_C = \frac{\text{MSC}}{\text{MSE}} = \frac{31.6667}{5.8333} \approx 5.43$$

with $(\nu_C, \nu_E) = (2, 12)$ d.f.

Step 8: ANOVA table and decision

Source	SS	d.f.	MS	F_{cal}	$F_{0.05}(\nu, \nu_E)$	Decision
Columns (treatments)	63.3333	2	31.6667	5.43	3.89	Significant
Error (within)	70	12	5.8333	–	–	–
Total	133.3333	14				

- $F_C = 5.43 > F_{0.05}(2, 12) = 3.89 \Rightarrow H_0^{(C)}$ is rejected: the difference between the three makes is **significant**.

Final Answer

The three makes of computers differ significantly in durability (frequency of repairs).

Question 4: Varietal Trial on Wheat

Question

A varietal trial was conducted on wheat with four varieties A, B, C, D in a Latin square design. The plan and observed yield per plot are given:

	Col 1	Col 2	Col 3	Col 4
Row 1	C: 25	B: 23	A: 20	D: 20
Row 2	A: 19	D: 19	C: 21	B: 18
Row 3	B: 19	A: 14	D: 17	C: 20
Row 4	D: 17	C: 20	B: 21	A: 15

Analyse the data and interpret the results at 5% LOS.

Answer

Step 1: Hypotheses

- $H_0^{(R)}$: there is no significant difference between the rows;
  $H_1^{(R)}$: there is a significant difference between the rows.
- $H_0^{(C)}$: there is no significant difference between the columns;
  $H_1^{(C)}$: there is a significant difference between the columns.
- $H_0^{(Tr)}$: there is no significant difference between the varieties;
  $H_1^{(Tr)}$: there is a significant difference between the varieties.

Step 2: Data table with row, column and treatment totals

Row	Col 1	Col 2	Col 3	Col 4	R_i (row total)
Row 1	C: 25	B: 23	A: 20	D: 20	88
Row 2	A: 19	D: 19	C: 21	B: 18	77
Row 3	B: 19	A: 14	D: 17	C: 20	70
Row 4	D: 17	C: 20	B: 21	A: 15	73
C_j (column total)	80	76	79	73	$T = 308$

Check: $\sum R_i = \sum C_j = T = 308$, order $n = 4$, $N = n^2 = 16$.
Collecting the observations of each treatment (letter):

Treatment	Observations				T_k (treatment total)
A	20	19	14	15	$T_A = 68$
B	23	18	19	21	$T_B = 81$
C	25	21	20	20	$T_C = 86$
D	20	19	17	17	$T_D = 73$
			Check: $\sum T_k$		$T = 308$

Step 3: Table of squares x_{ij}^2
Squaring every observation:

Row	Col 1	Col 2	Col 3	Col 4	$\sum_j x_{ij}^2$
Row 1	625	529	400	400	1954
Row 2	361	361	441	324	1487
Row 3	361	196	289	400	1246
Row 4	289	400	441	225	1355
$\sum_i x_{ij}^2$	1636	1486	1571	1349	$\sum\sum x_{ij}^2 = 6042$

Step 4: Correction factor

$$\text{CF} = \frac{T^2}{N} = \frac{308^2}{16} = \frac{94864}{16} = 5929$$

Step 5: Sums of squares
Here $\text{SSR} = \frac{1}{n}\sum R_i^2 - \text{CF}$, $\text{SSC} = \frac{1}{n}\sum C_j^2 - \text{CF}$, $\text{SSTr} = \frac{1}{n}\sum T_k^2 - \text{CF}$.

$$\text{SST} = \sum\sum x_{ij}^2 - \text{CF} = 6042 - 5929 = 113$$

$$\begin{aligned}\text{SSR} &= \frac{88^2 + 77^2 + 70^2 + 73^2}{4} - \text{CF} \\ &= \frac{23902}{4} - 5929 = 5975.5 - 5929 = 46.5\end{aligned}$$

$$\begin{aligned}\text{SSC} &= \frac{80^2 + 76^2 + 79^2 + 73^2}{4} - \text{CF} \\ &= \frac{23746}{4} - 5929 = 5936.5 - 5929 = 7.5\end{aligned}$$

$$\begin{aligned}\text{SSTr} &= \frac{68^2 + 81^2 + 86^2 + 73^2}{4} - \text{CF} \\ &= \frac{23910}{4} - 5929 = 5977.5 - 5929 = 48.5\end{aligned}$$

$$\begin{aligned}\text{SSE} &= \text{SST} - \text{SSR} - \text{SSC} - \text{SSTr} \\ &= 113 - 46.5 - 7.5 - 48.5 = 10.5\end{aligned}$$

Step 6: Degrees of freedom and mean squares

$\nu_R = \nu_C = \nu_{Tr} = n - 1 = 3, \quad \nu_E = (n-1)(n-2) = 6, \quad \nu_T = n^2 - 1 = 15.$

$$\text{MSR} = \frac{\text{SSR}}{\nu_R} = \frac{46.5}{3} = 15.5$$

$$\text{MSC} = \frac{\text{SSC}}{\nu_C} = \frac{7.5}{3} = 2.5$$

$$\text{MSTr} = \frac{\text{SSTr}}{\nu_{Tr}} = \frac{48.5}{3} \approx 16.1667$$

$$\text{MSE} = \frac{\text{SSE}}{\nu_E} = \frac{10.5}{6} = 1.75$$

Step 7: F-ratios

$$F_R = \frac{\text{MSR}}{\text{MSE}} = \frac{15.5}{1.75} \approx 8.86$$

$$F_C = \frac{\text{MSC}}{\text{MSE}} = \frac{2.5}{1.75} \approx 1.43$$

$$F_{Tr} = \frac{\text{MSTr}}{\text{MSE}} = \frac{16.1667}{1.75} \approx 9.24$$

All three ratios have $(n-1, (n-1)(n-2)) = (3, 6)$ d.f.

Step 8: ANOVA table and decision

Source	SS	d.f.	MS	F_{cal}	$F_{0.05}(\nu, \nu_E)$	Decision
Rows	46.5	3	15.5	8.86	4.76	Significant
Columns	7.5	3	2.5	1.43	4.76	Not significant
Treatments	48.5	3	16.1667	9.24	4.76	Significant
Error	10.5	6	1.75	–	–	–
Total	113	15				

- $F_R = 8.86 > F_{0.05}(3,6) = 4.76 \Rightarrow H_0^{(R)}$ is rejected: the difference between the rows is **significant**.
- $F_C = 1.43 < F_{0.05}(3,6) = 4.76 \Rightarrow H_0^{(C)}$ is accepted: the difference between the columns is **not significant**.
- $F_{Tr} = 9.24 > F_{0.05}(3,6) = 4.76 \Rightarrow H_0^{(Tr)}$ is rejected: the difference between the varieties is **significant**.

Final Answer

The wheat varieties differ significantly in yield, and so do the rows; the column (second blocking factor) effect is not significant.